DM548: Computer Architecture & System Programming
DM588: Computer Architecture

Fall 2026

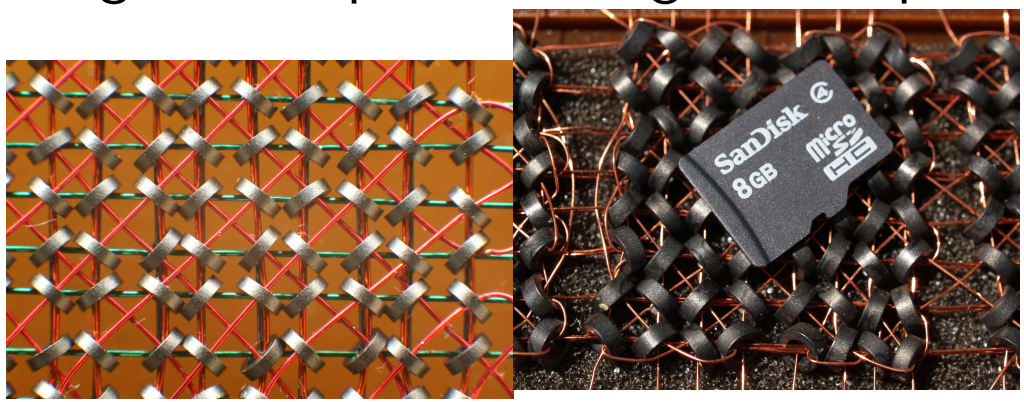
Internal and External Memory

- Internal Memory
  - General
  - DDR SDRAM
  - eDRAM
  - Flash Memory
  - Newer Non-volatile Memory Technologies
- External Memory
  - Magnetic Disks
  - RAID
  - Solid State Drives
  - Optical Discs
- Error Correction Codes

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Internal Memory

- In the past, main memory employed an array of doughnut-shaped ferromagnetic loops, referred to as cores.



8 B vs. 8 GB

- Hence, main memory was often referred to as core.
- This is also where the term core dump originates from.
- Today, the use of semiconductor chips for main memory is almost universal.
- The basic element is the memory cell:
  - A variety of electronic technologies are used.
  - They exhibit two stable (or semistable) states: binary 0 and 1.
  - They are capable of being written into (at least once), to set the state.
  - They are capable of being read to sense the state.

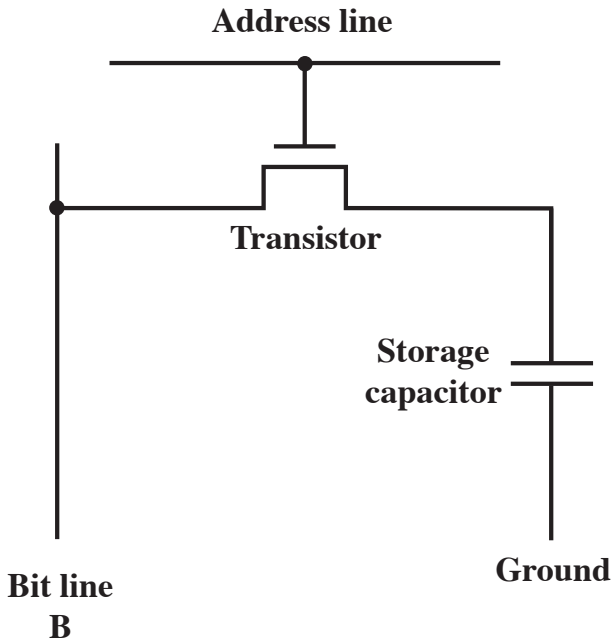
Memory Types

<table>
<tr><th>Memory type</th><th>Category</th><th>Erasure</th><th>Write Mechanism</th><th>Volatility</th></tr>
<tr><td>Random-access memory (RAM)</td><td>Read-write memory</td><td>Electrically, byte-level</td><td>Electrically</td><td>Volatile</td></tr>
<tr><td>Read-only memory (ROM)</td><td rowspan="2">Read-only memory</td><td rowspan="2">Not possible</td><td>Masks</td><td rowspan="4">Non-volatile</td></tr>
<tr><td>Programmable ROM (PROM)</td><td rowspan="3">Electrically</td></tr>
<tr><td>Erasable PROM (EPROM)</td><td rowspan="3">Read-mostly memory</td><td>UV light, chip-level</td></tr>
<tr><td>Electrically EPROM (EEPROM)</td><td>Electrically, byte-level</td></tr>
<tr><td>Flash memory</td><td>Electrically, block-level</td></tr>
</table>

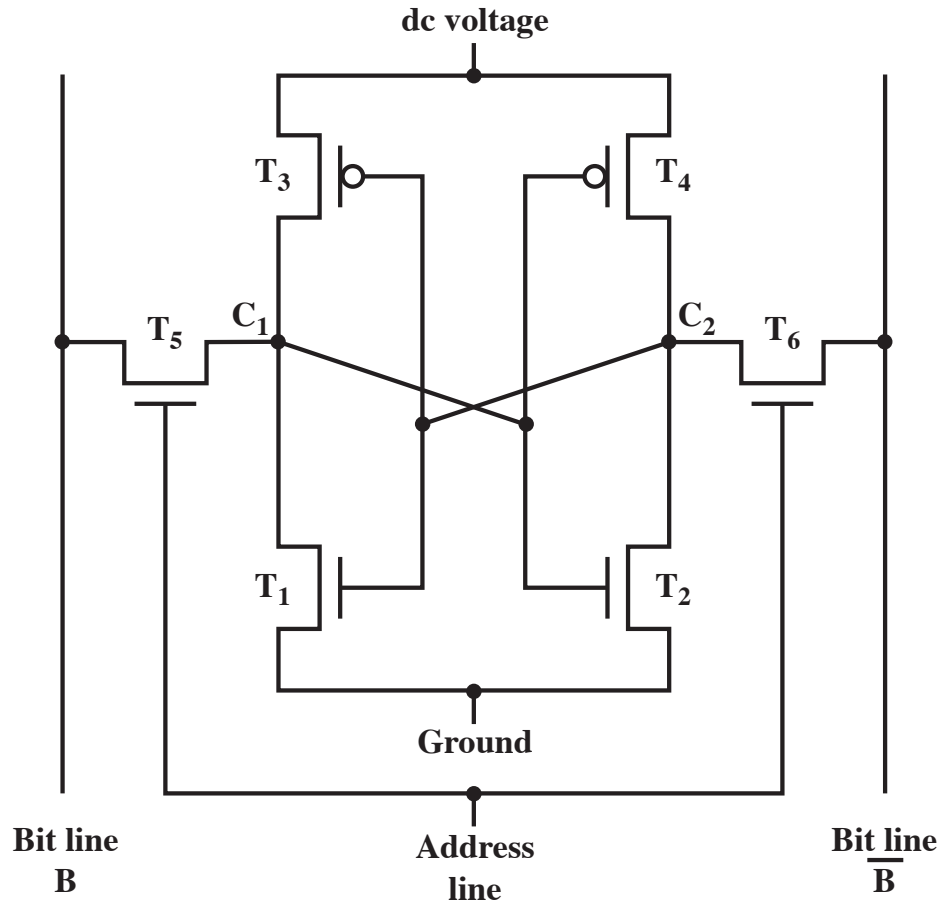
Two Different RAM Technologies

- Dynamic RAM (DRAM)
  - Made with cells that sore data as charge on capacitors.
  - Absence or presence of charge in a capacitor is interpreted as binary 0 or 1.
  - Requires periodic charge refreshing to maintain data storage.
  - The term dynamic refers to tendency of the stored charge to leak away, even with power continuously applied.
- Static RAM (SRAM)
  - Digital device that uses the same logic elements used in processors.
  - Binary values are stored using traditional flip-flop logic gate configurations.
  - Will hold its data as long as power is supplied to it.

DRAM and SRAM



(a) Dynamic RAM (DRAM) cell



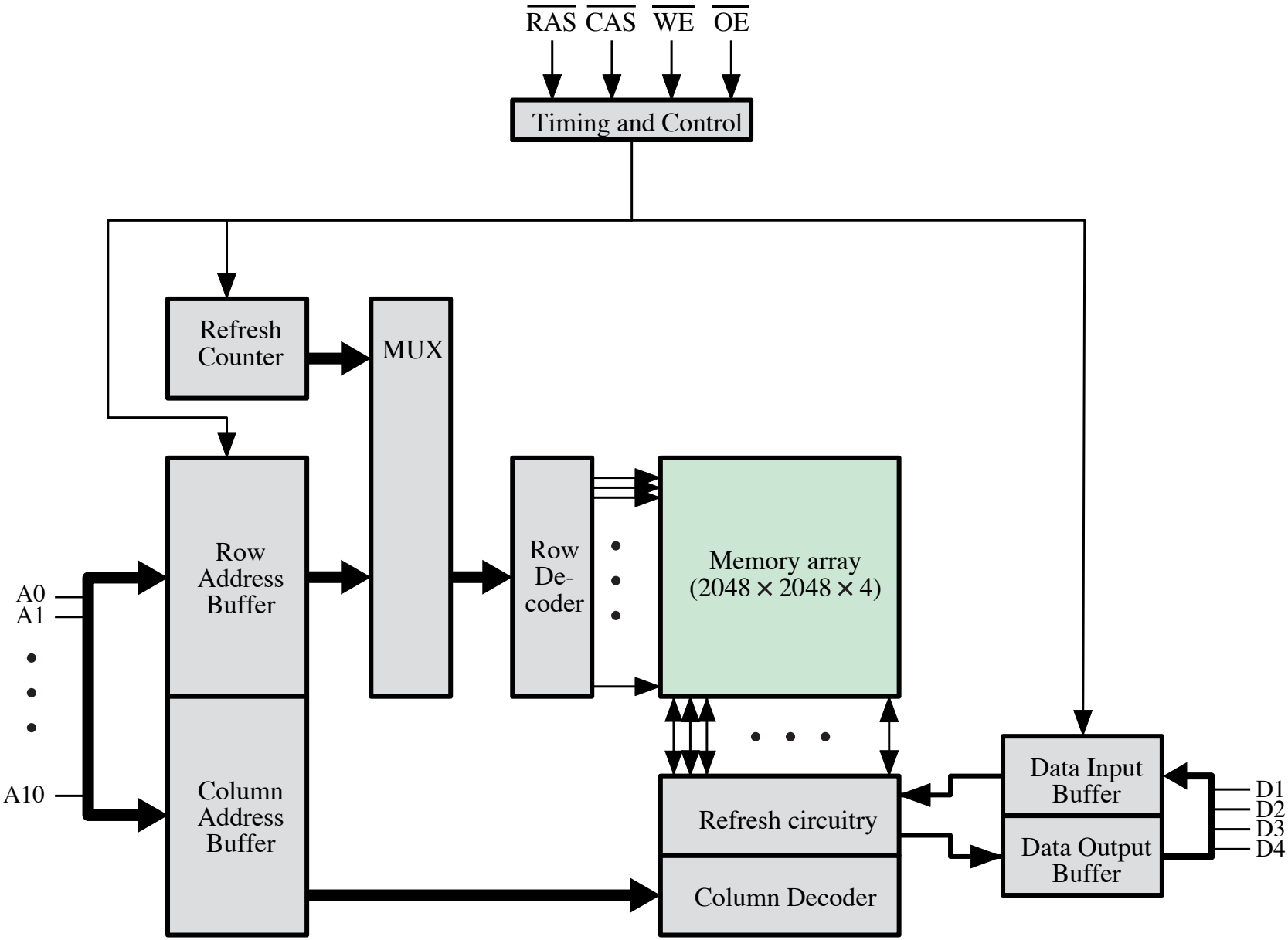
(b) Static RAM (SRAM) cell

Figure 6.2 Typical Memory Cell Structures

SRAM vs. DRAM

- Both are volatile
  - Power must be continuously supplied to preserve the bit values.
- Dynamic cell
  - Simpler to build, is smaller
  - Denser packing and less expensive
  - Requires supporting refresh circuitry
  - Tend to be favoured for large memory requirements
  - Used for main memory
- Static
  - Faster
  - Used for cache memory (both on and off chip)

Schematic of a Typical DRAM Array



Ranks: Multiple Arrays as One, Parallelized Read/Write

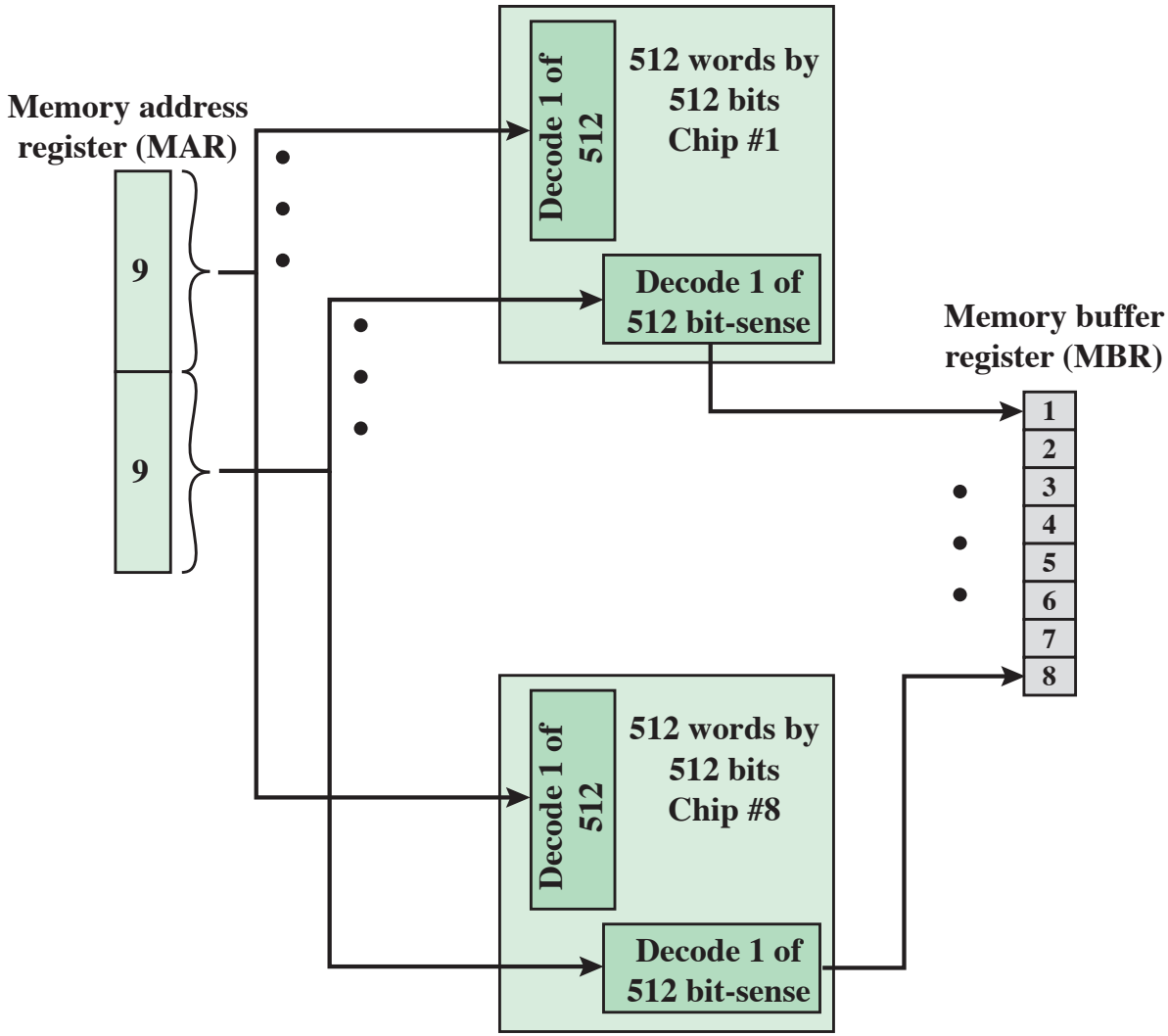
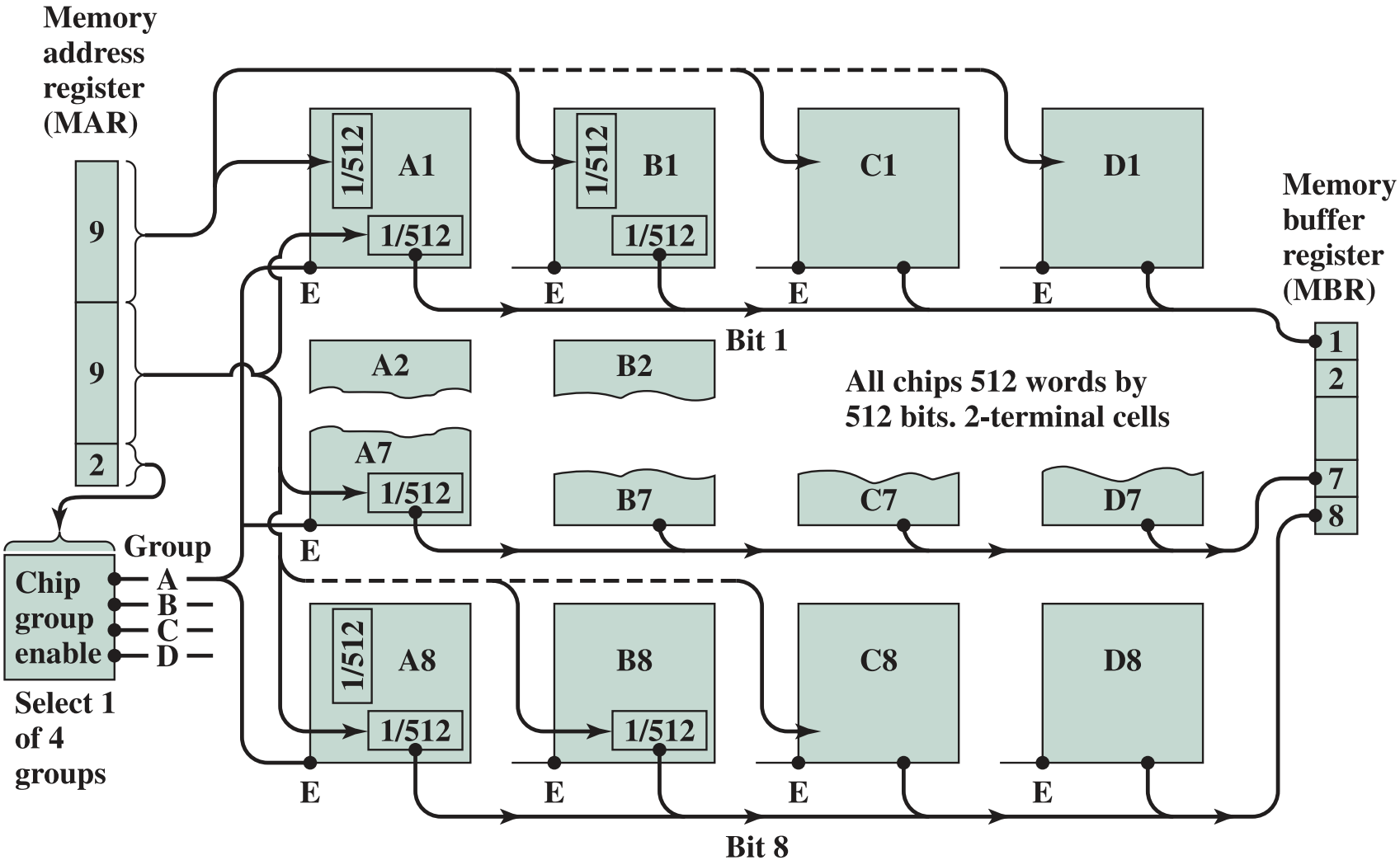


Figure 6.5 256-KByte Memory Organization

Multiple Ranks



- Two address bits are used to select a group (A, B, C, or D).
- Within a group the arrays in parallel retrieves data.

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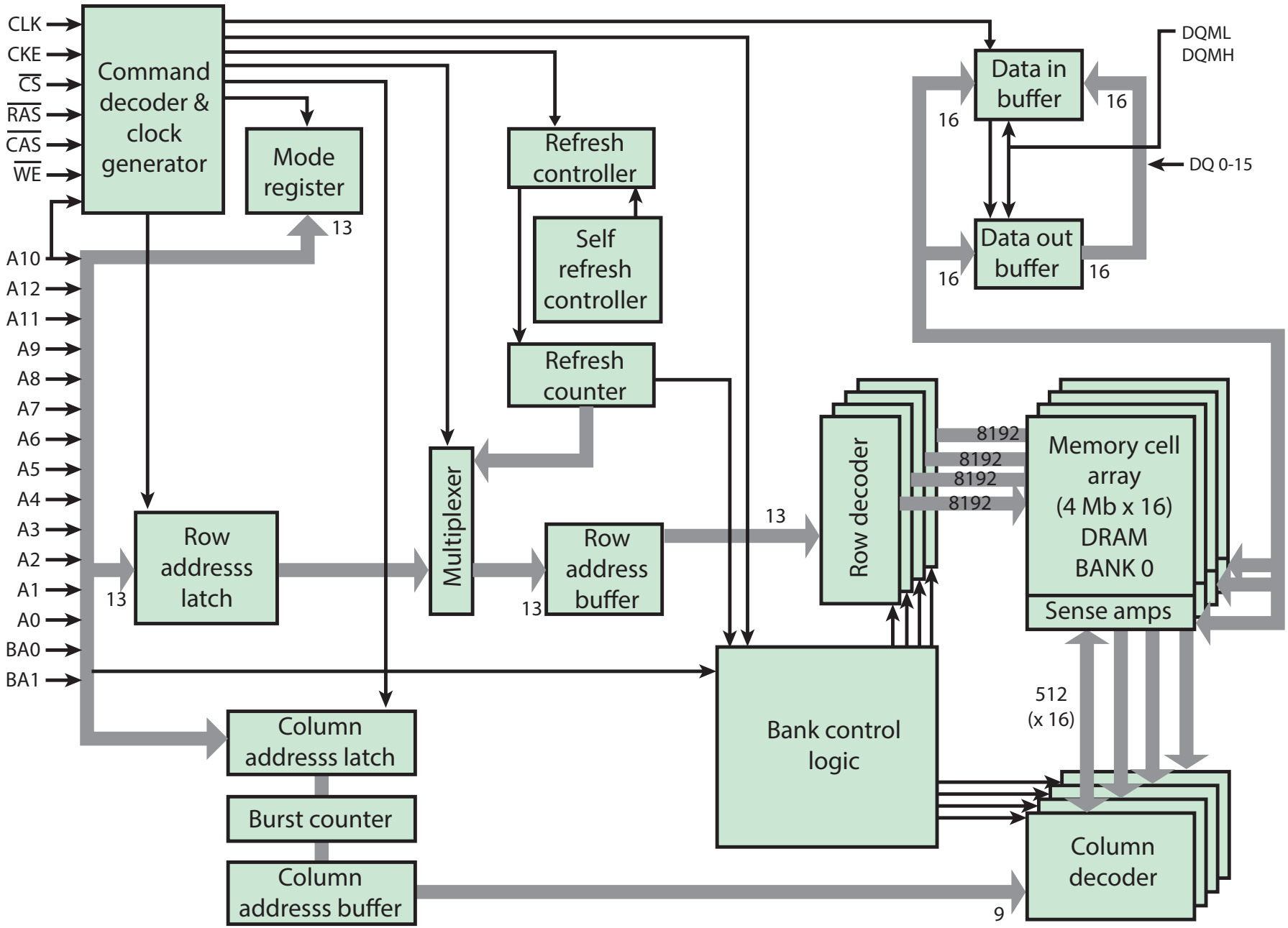
DRAM Organization

- One of the most critical system bottlenecks is the access to main memory.
- Static RAM is too expensive for implementing main memory.
- The traditional DRAM chip is the dominating implementation.
- The internals of DRAM chips has not changed much since the 1970's, until recently.

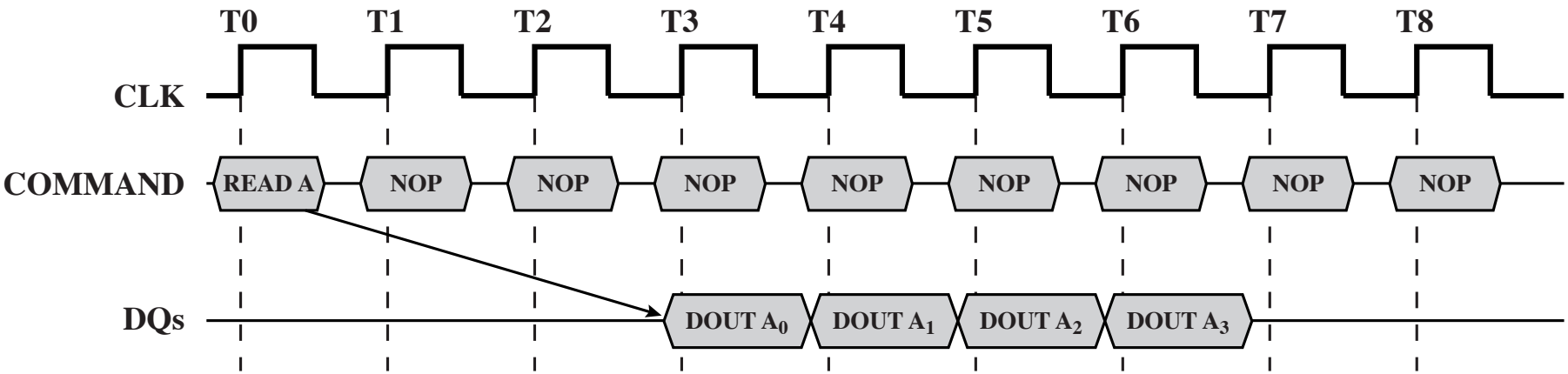
Synchronous DRAM (SDRAM)

- The most widely used form of DRAM.
- Data exchange with the system is synchronized by an external clock signal (from the CPU or memory bus).
- The memory controller issues an instruction to the SDRAM which latches command, address information, and potential data to be written.
- After a specific delay, the access time, the data is written or read.

SDRAM Module



SDRAM Read Timing



SDRAM Read Timing (Burst Length: 4, $\overline{\text{CAS}}$ latency: 2)

- Opening a row takes most of the time.
- Once a row is open, column access is fast.
- The SDRAM can then burst out multiple data words.
- The minimum burst length is also called the "prefetch" length.

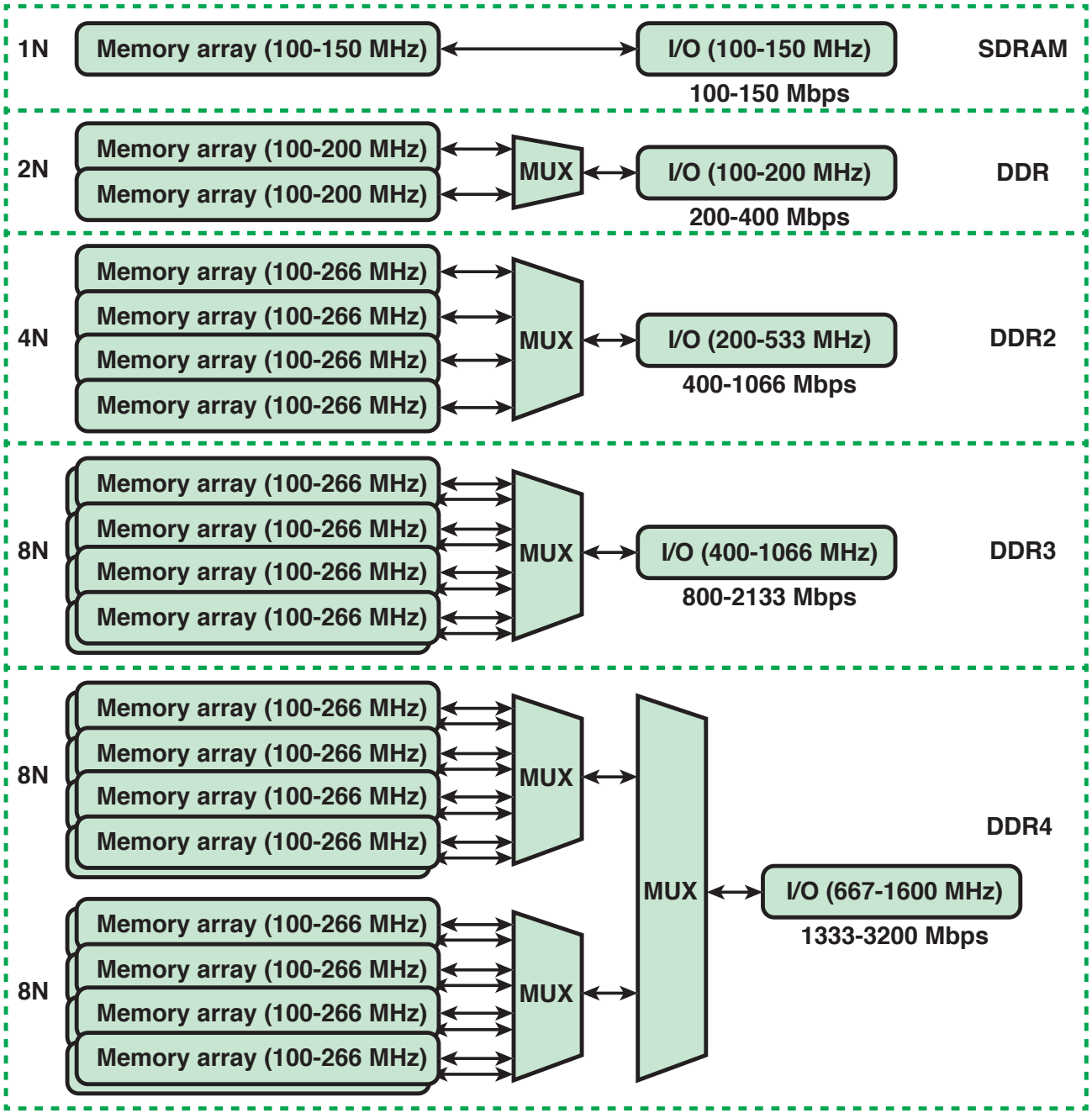
Double Data Rate SDRAM (DDR SDRAM)

- The DRAM it self can not currently be made significantly faster.
- The previous SDRAM could only send data once per clock cycle.
- DDR SDRAM sends data twice per clock cycle, on the rising and on the falling edge.
- DDR developed as series of standards by the JEDEC Solid State Technology Association (Joint Electron Device Engineering Council).

Type	Year	Int. Freq. (MHz)	Prefetch	Bus Freq. (MHz)	Bandwidth (GB/s)
SDR	1993	66–200	1n	66–200?	1.056–1.6?
DDR	2000	100–200	2n	100–200	1.6–3.2
DDR2	2003	100–266.6	4n	200–533.3	3.2–8.533
DDR3	2007	100–266.6	8n	400–1066.6	6.4–17.066
DDR4	2014	200–400	8n	800–1600	12.8–25.6
DDR5	2021	200–400	16n	1600–3200	25.6–51.2

`https://en.wikipedia.org/wiki/DDR_SDRAM`

DDR Generations



DDR Memory Organization

- The CPU may have one or more independent channels.
- To each channel a number of DIMMs (Dual Inline Memory Module), i.e, a number of the physical things we can buy.
- Each DIMM contains a number of ranks.
- Each rank usually contains 8 (or 9 for ECC (see later)) chips.
- Each chip has multiple arrays.
- Arrays across a rank form a bank.
- Each array has a number of rows, of which at most 1 can be open for reading/writing at a time.
- When reading/writing, a rank works as one entity, so a single 64 bit read consists of 8 bits from each of the 8 chips in a rank.
- With an 8n prefetch, a single read command will thus extract $8 \cdot 64\,\text{bit} = 64\,\text{B}$, i.e., a complete cache line on x86!
- (In DDR5 each DIMM provides two channels, each only 32 bits wide, and then the prefetch is doubled, to still provide 64 B per read command.)

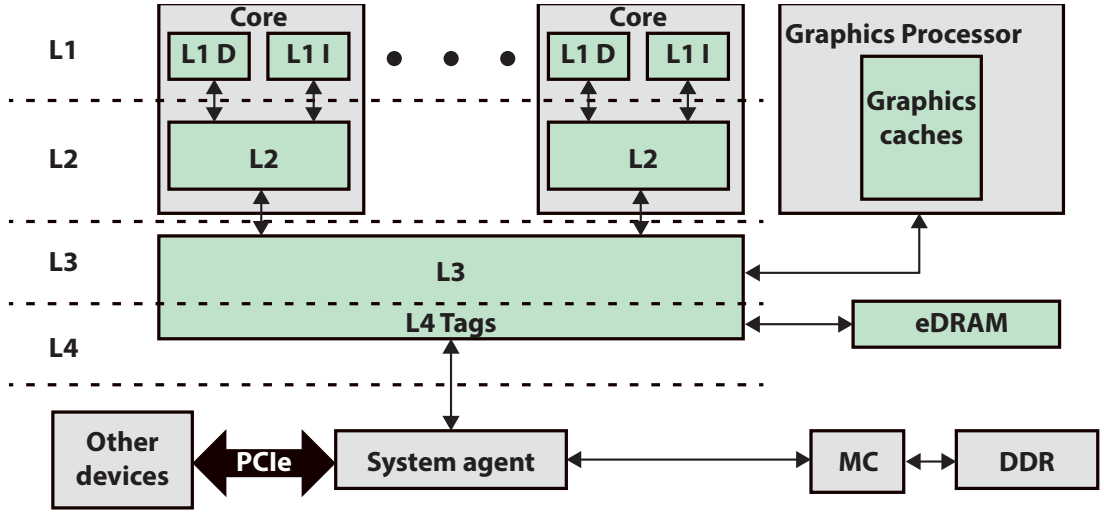
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eDRAM, Embedded DRAM

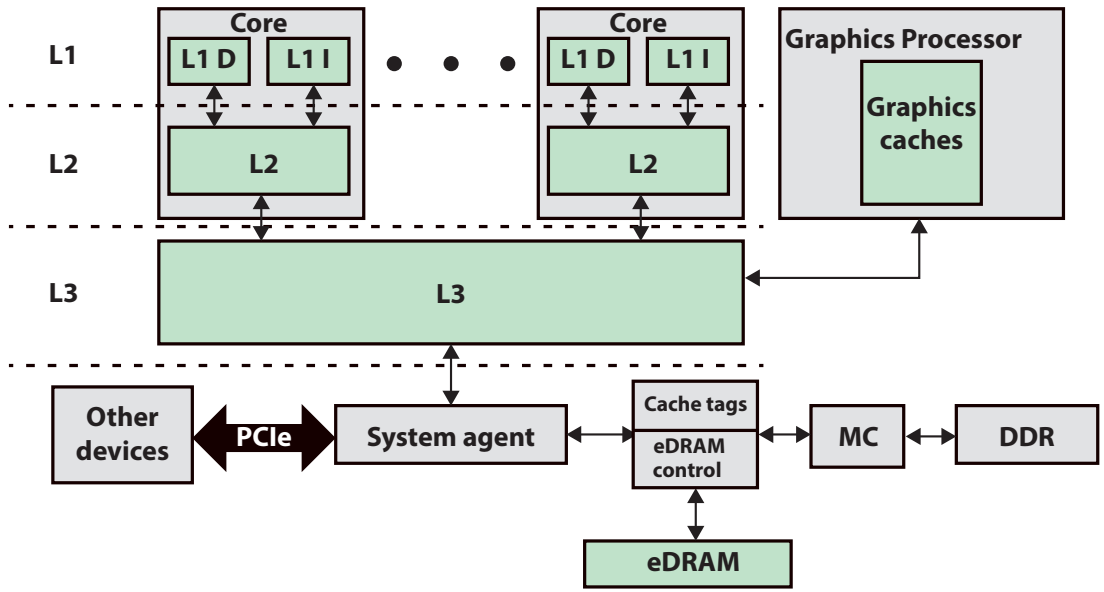
Put DRAM in the same package as the CPU:

- space-wise better than SRAM, but worse than ordinary DRAM,
- cheaper than SRAM, but more expensive than ordinary DRAM,
- slower access than SRAM, but faster than ordinary DRAM, due to proximity,
- generally less power drain than SRAM.

Example: eDRAM in Intel Core CPUs



(a) Original use of eDRAM



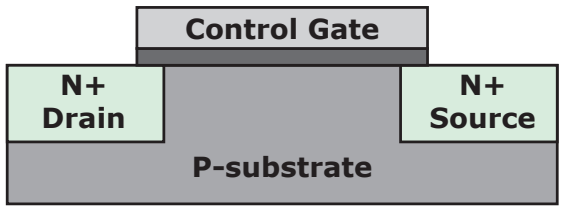
(b) More recent use of eDRAM

MC = memory controller

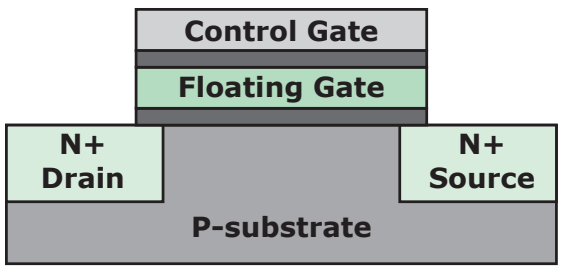
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Flash Memory

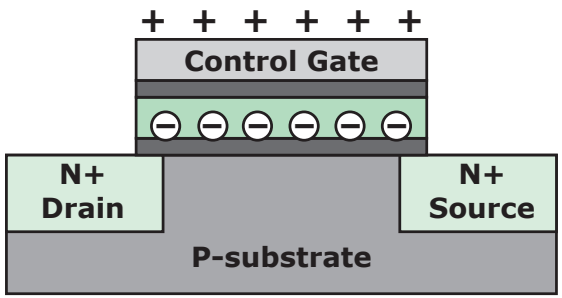
- Non-volatile.
- Can be used both as internal and external memory.
- Applying a large voltage traps/releases electrons in the floating gate.
- Binary 0 and 1 appears by testing if it works as a transistor or not, using a small voltage.



(a) Transistor structure



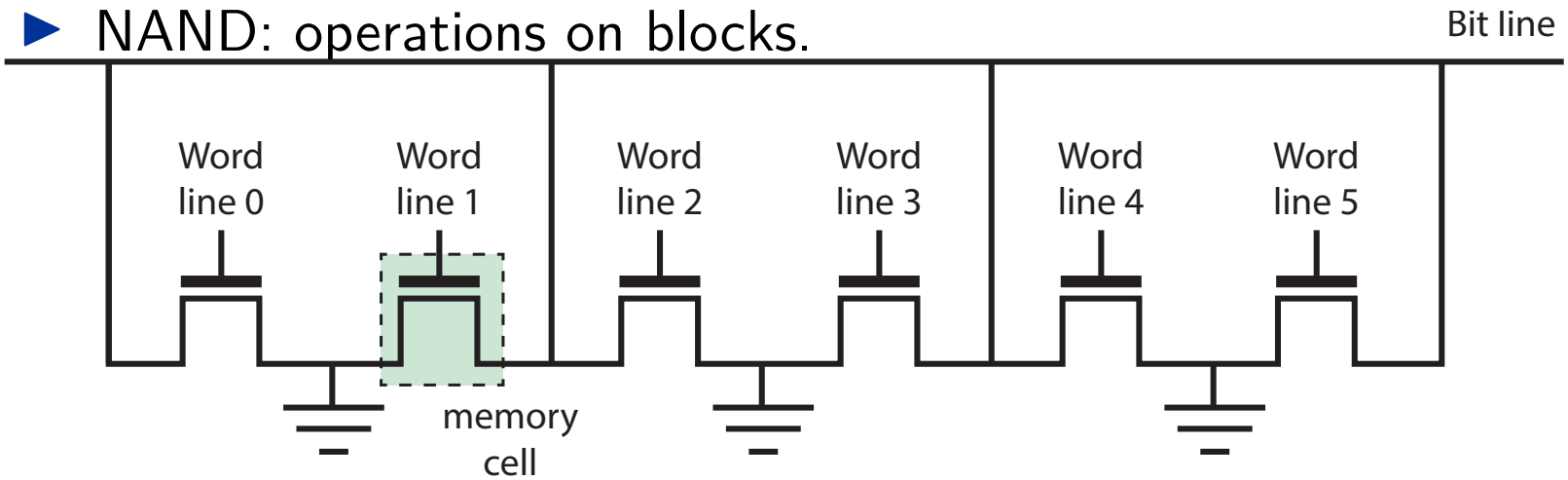
(b) Flash memory cell in one state



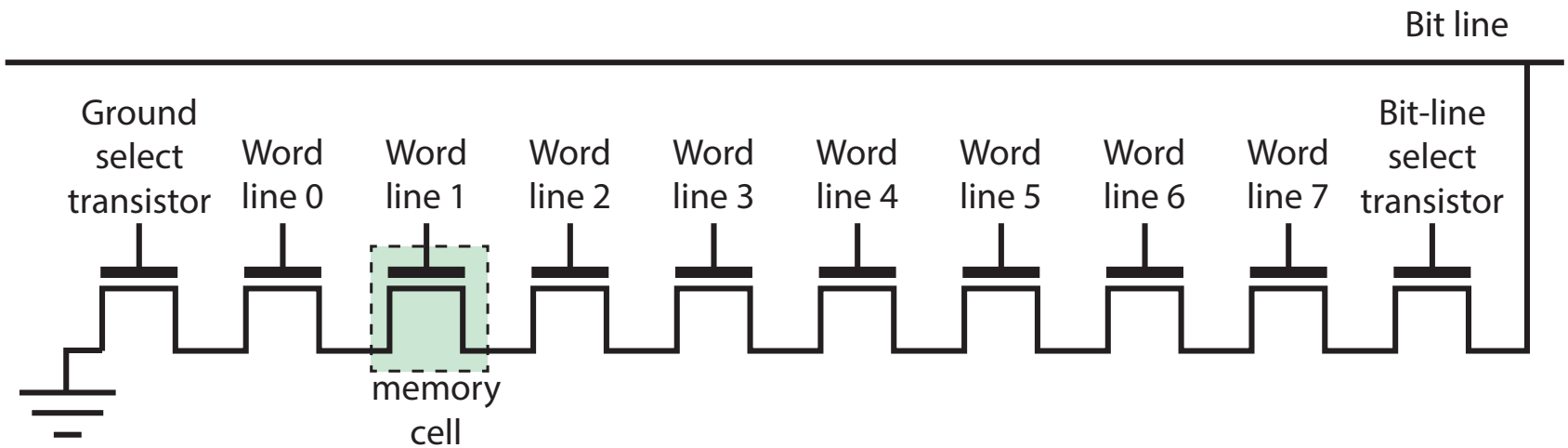
(c) Flash memory cell in zero state

Flash Memory Organization: NOR and NAND

- NOR: operations on bits/bytes.
- NAND: operations on blocks.



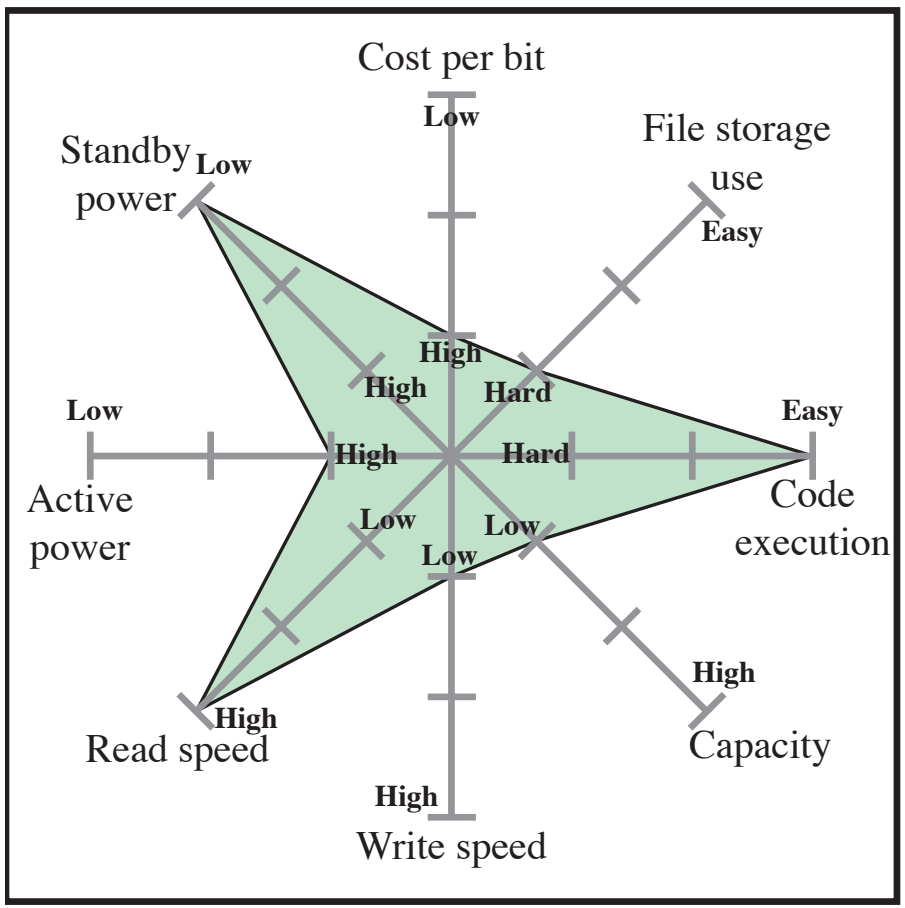
(a) NOR flash structure



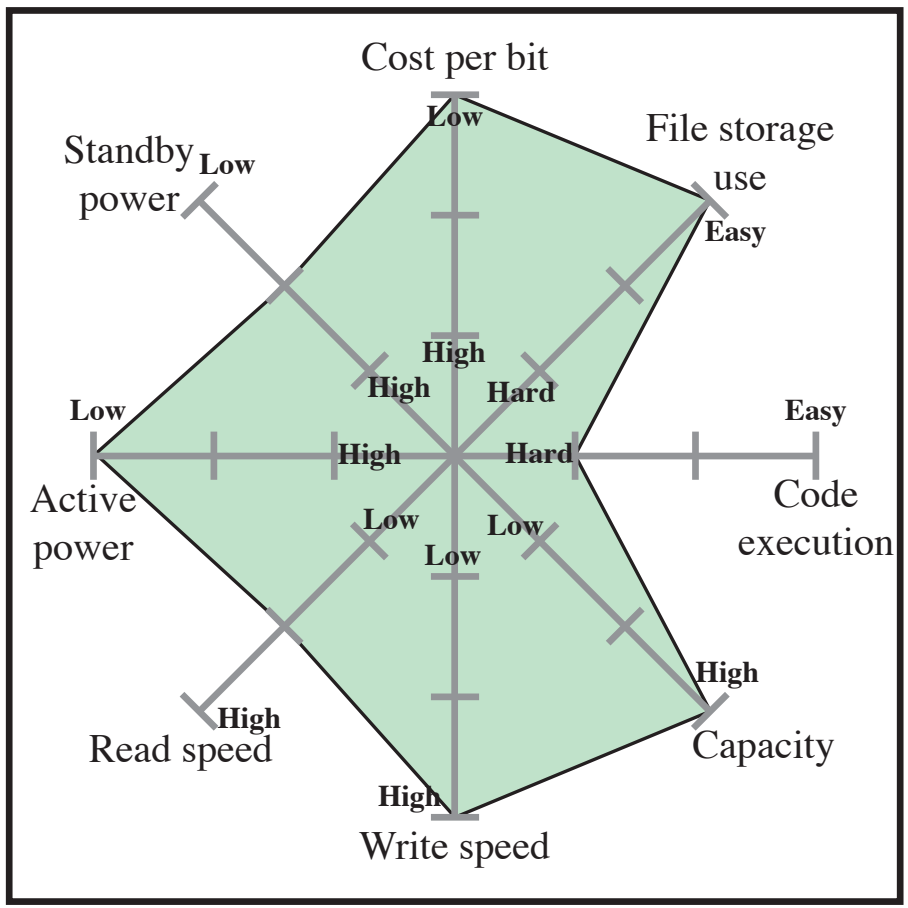
(b) NAND flash structure

Flash Memory Organization: NOR and NAND

- NOR: favoured in internal memory.
- NAND: favoured in external memory.



(a) NOR



(b) NAND

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STT-RAM, PCRAM, and ReRAM

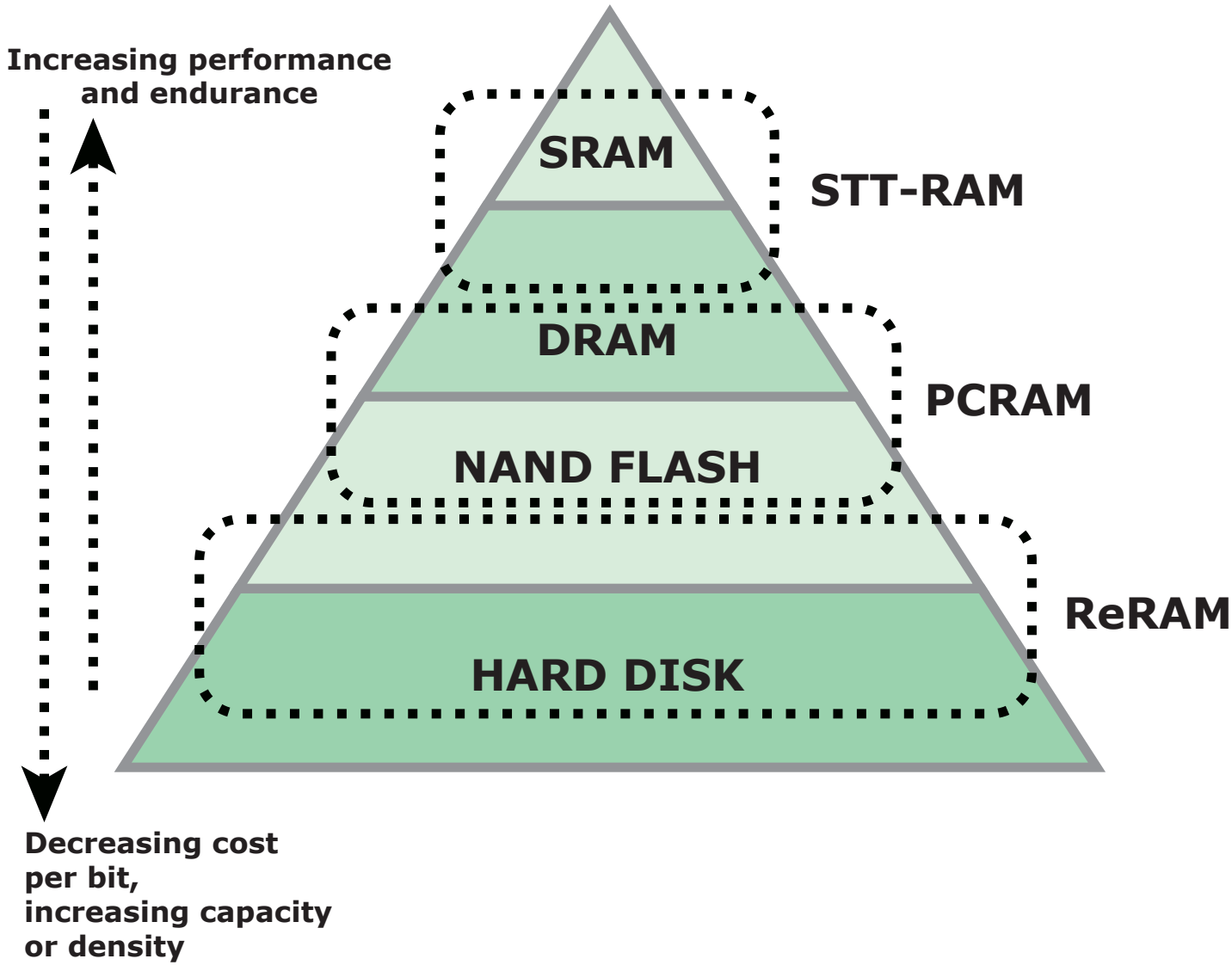


Figure 6.20 Nonvolatile RAM within the Memory Hierarchy

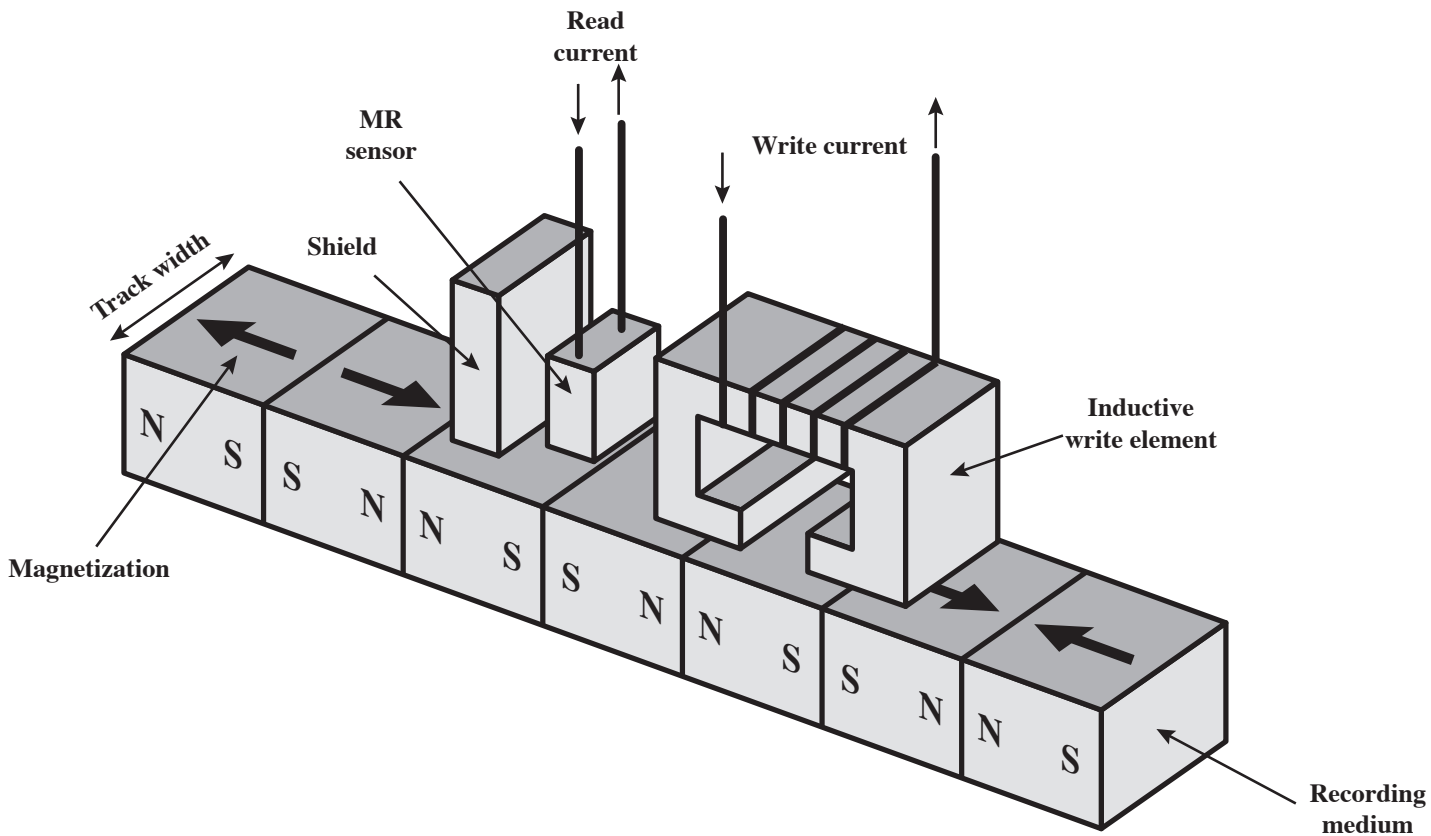
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Magnetic Disks

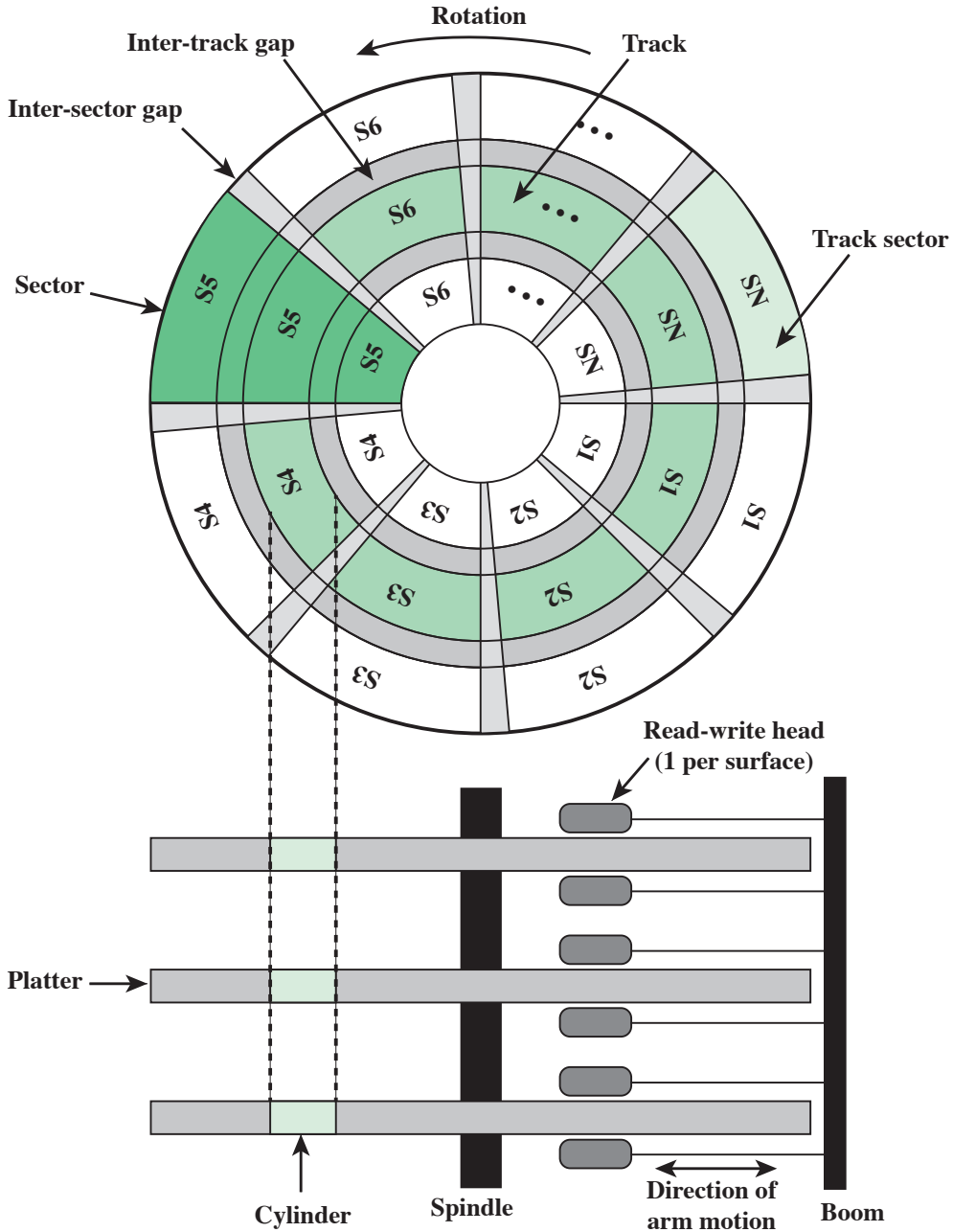
- A non-magnetic material, the substrate, coated with a magnetizable material.
- The substrate is typically aluminium, an aluminium alloy, or sometimes glass.
- Benefits of glass substrate:
  - Improved uniformity of the magnetic film surface, increasing reliability.
  - Significant reduction in overall surface defects, reducing read/write errors.
  - Ability to support lower fly heights.
  - Better stiffness, reducing disk dynamics.
  - Greater ability to withstand shock and damage.

The Read/Write Head

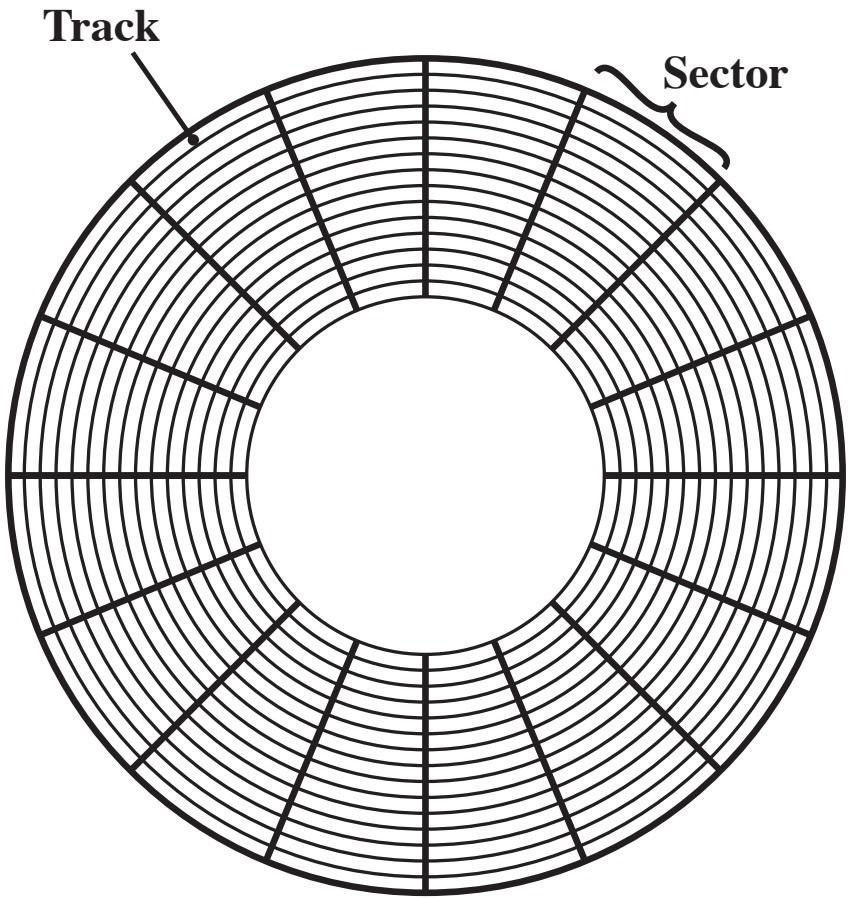
- Data are written and read from the disk via a conducting coil named the head.
- During a read or write operation the head is stationary while the platter rotates beneath it.
- The write mechanism exploits the fact that electricity flowing through a coil produces a magnetic field.



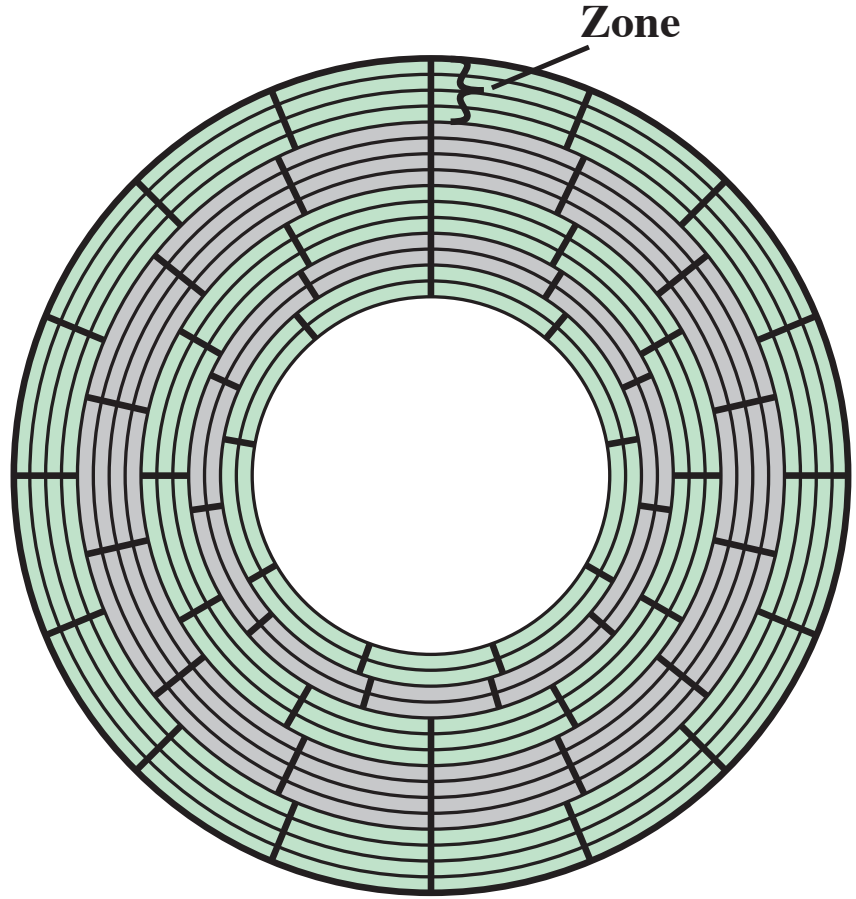
General Disk Layout



Disk Layout Methods

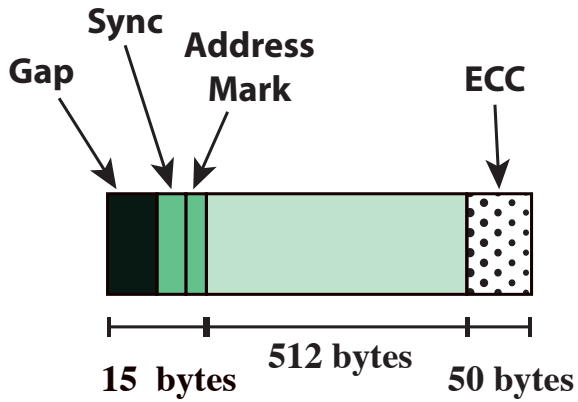


(a) Constant angular velocity

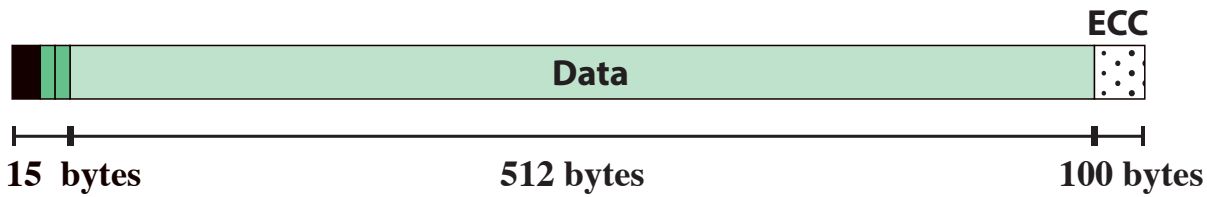


(b) Multiple zone recording

Winchester Disk Format



(a) Legacy 512-byte sector



(b) Advanced Format 4k-byte sector

Note: book error in the caption of (b), the data is 4096 bytes long.

Physical Characteristics of Disk Systems

- Head motion
  - Fixed head (one per track)
  - Movable head (one per surface)
- Disk portability
  - Non-removable disk
  - Removable disk (e.g., floppy disks)
- Sides
  - Single sided
  - Double sided
- Platters
  - Single platter
  - Multiple platter
- Head Mechanism
  - Contact (e.g., floppy disk)
  - Fixed gap
  - Aerodynamic gap (Winchester)

Head Mechanisms

- The head must generate or sense an electromagnetic field of sufficient magnitude to write and read properly.
- A narrower head requires smaller gap to the platter surface.
- A narrower head gives narrower tracks, and thereby higher data density.
- But a smaller gap gives a higher risk of error due to impurities or imperfections.

Winchester head

- Used in sealed drive assemblies that are almost free of contaminants.
- Designated to operate closer to the surface of the disk than conventional rigid headers.
- The head is a small airfoil:
  - When the disk is still, it rests on the platter surface.
  - Air pressure from the spinning platter is enough to make the head fly.

Disk Performance Parameters

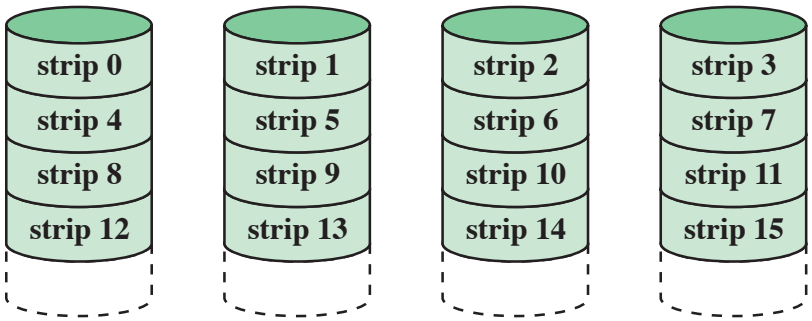
- The disks are rotating at a constant speed.
- To read/write:
  - The heads must be moved to the desired track mechanically.
  - The right platter side must be selected electronically.
  - We must wait for the platter to rotate so the the desired sector is lined up.
- Seek time: the time it takes to position the head at the track.
- Rotational delay/latency: the time it takes until the beginning of the sector is lined up.
- Access time: seek time + rotational delay
- Transfer time: the time it takes to perform the read/write operation once the sector is lined up, while the platter rotates beneath the head.

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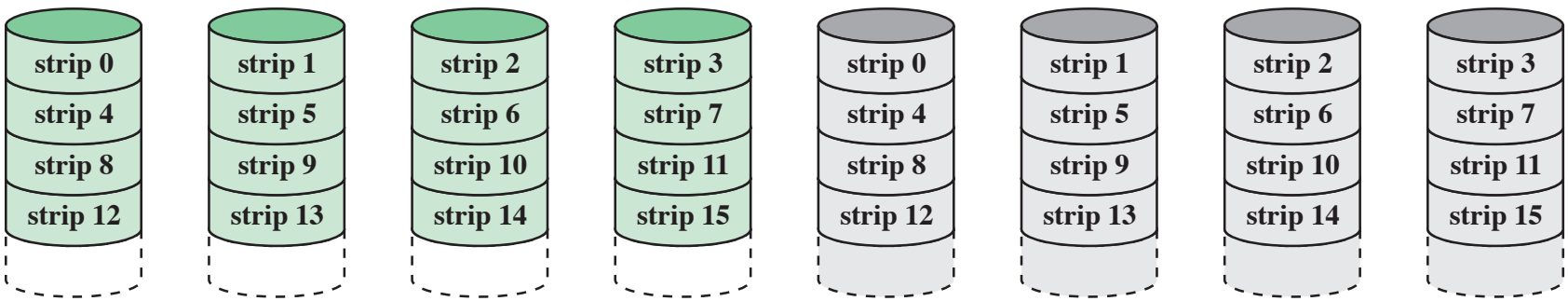
Overview

- Redundant Array of Inexpensive/Independent Disks/Drives
- There are 7 different levels of RAID.
- "Level" is misleading, there is no hierarchy, rather think "configuration".
- Make a set of physical disk drives look as a single logical drive to the OS.
- Data is distributed across the disks, using striping, mirroring, and parity.
- Redundant disk capacity is used to store extra data, which guarantees data recoverability is case of disk failure.

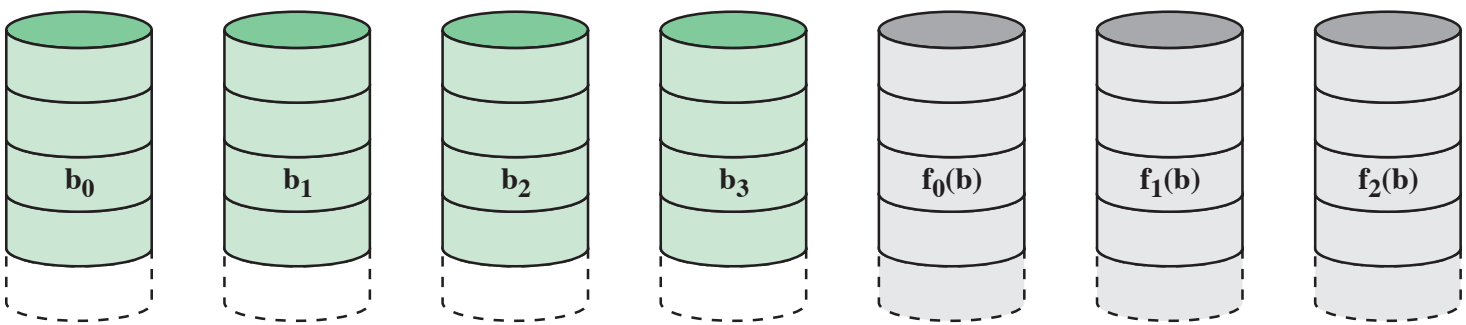
RAID 0, 1, 2



(a) RAID 0 (non-redundant)

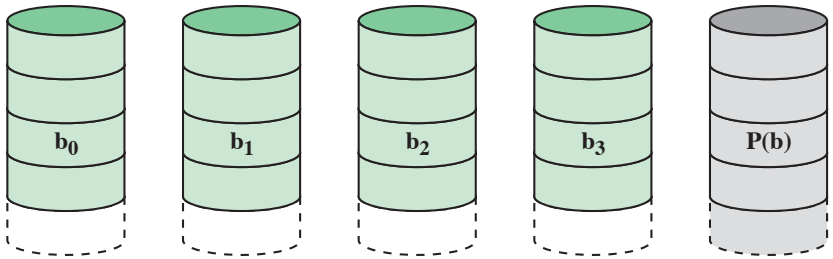


(b) RAID 1 (mirrored)

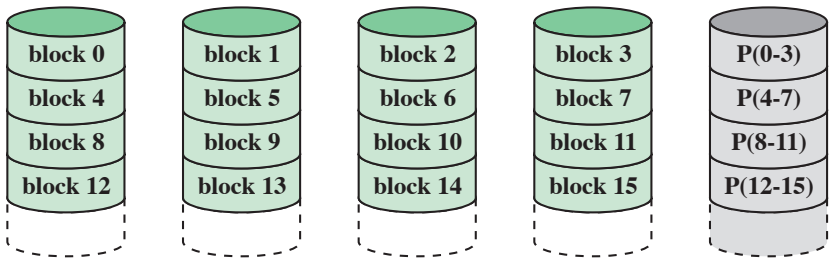


(c) RAID 2 (redundancy through Hamming code)

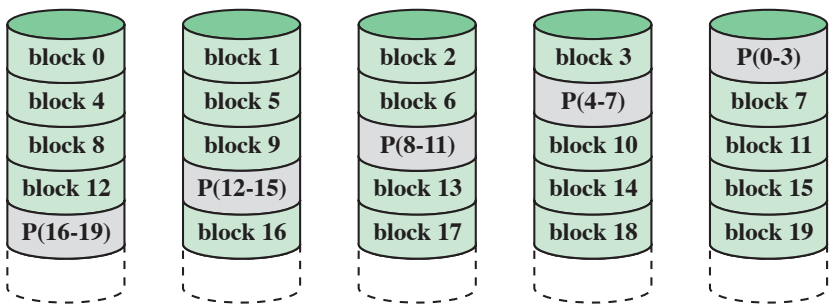
RAID 3–6



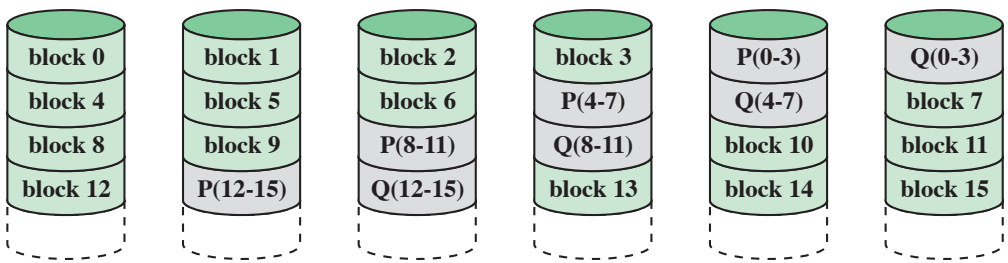
(d) RAID 3 (bit-interleaved parity)



(e) RAID 4 (block-level parity)



(f) RAID 5 (block-level distributed parity)



(g) RAID 6 (dual redundancy)

RAID Overview, Levels 0, 1, 2

L	Advantages	Disadvantages	Applications
0	▶ I/O performance greatly improved, load is spread across many channels and drives. ▶ No parity calculation overhead. ▶ Very simple design. ▶ Easy to implement.	▶ The failure of just one drive will result in all data in an array being lost.	▶ Video production and editing ▶ Image editing ▶ Prepress applications ▶ Any application requiring high bandwidth
1	▶ 100 % redundancy of data, no rebuild necessary in case of disk failure, just a copy to the replacement disk. ▶ Can sustain multiple simultaneous drive failures in some cases. ▶ Simplest RAID design.	▶ Highest disk overhead of all RAID types.	▶ Accounting ▶ Payroll ▶ Financial ▶ Any application requiring very high availability.
2	▶ Extremely high data transfer rate possible. ▶ The higher the transfer rate required, the better the ratio of disks to ECC disks. ▶ Relatively simple controller design, compared to level 3, 4, and 5.	▶ Very high ratio of ECC disks to data disks with smaller word sizes. ▶ Entry level cost very high, must have high transfer rate requirement to justify.	▶ No commercial implementations exist. Not commercially viable.

RAID Overview, Levels 3–6

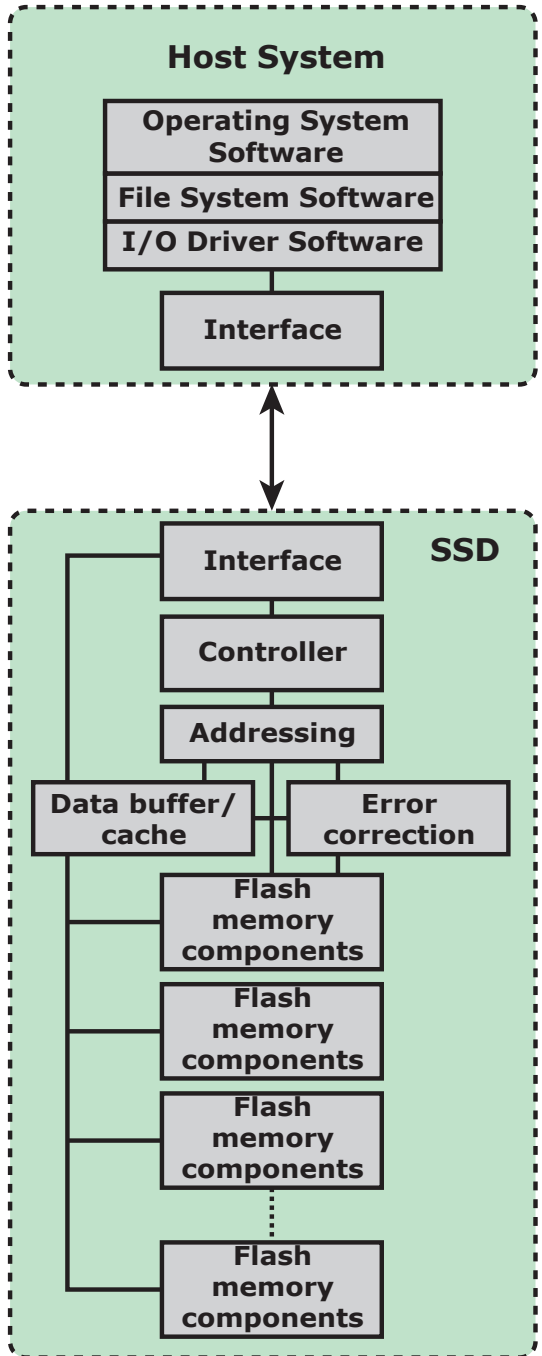
L	Advantages	Disadvantages	Applications
3	▶ Very high read/write transfer rate. ▶ Disk failure has insignificant impact on throughput. ▶ Low ratio of ECC (parity) disks to data disks.	▶ Transaction rate equal to that of a single disk drive at best, when spindles are synchronized. ▶ Controller design fairly complex.	▶ Video production and live-streaming ▶ Image editing ▶ Video editing ▶ Any application requiring high throughput
4	▶ Very high read transaction rate. ▶ Low ratio of ECC (parity) disks to data disks.	▶ Quite complex controller design. ▶ Worst write transaction rate and write aggregate rate. ▶ Difficult and inefficient data rebuild in the event of disk failure.	▶ No commercial implementations exist. Not commercially viable.
5	▶ Highest read transaction rate. ▶ Low ratio of ECC (parity) disks to data disks. ▶ Good aggregate transfer rate. ▶ Most versatile RAID level.	▶ Most complex controller design. ▶ Difficult to rebuild in the event of a disk failure, compared to level 1.	▶ File and application servers ▶ Database servers ▶ Web, email, and news servers ▶ Intranet servers
6	▶ Extremely high data fault tolerance. Can sustain multiple simultaneous drive failures.	▶ More complex controller design. ▶ Controller overhead to compute parity addresses is extremely high.	▶ Perfect solution for mission critical applications.

- Internal Memory
  - General
  - DDR SDRAM
  - eDRAM
  - Flash Memory
  - Newer Non-volatile Memory Technologies
- External Memory
  - Magnetic Disks
  - RAID
  - Solid State Drives
  - Optical Discs
- Error Correction Codes

Solid State Drives (SSDs)

- "Solid state" because they are circuitry built with semiconductors.
- Typically made with NAND flash memory.
- Complements and often replaces hard disk drives (HDDs).
- Advantages compared to HDDs:
  - High-performance input/output operations per second
  - Durability
  - Longer lifespan
  - Lower power consumption
  - Quieter and cooler running capabilities
  - Lower access times and latency rates
- NAND Flash Drive vs. Seagate Laptop Internal HDD:
  - File copy/write speed: 200–550 Mbps vs. 50–120 Mbps
  - Power draw/battery life: avg. 2–3 W vs. avg. 6–7 W
  - Storage capacity: up to 1 TB vs. up to 4 TB
  - Cost: ∼\$0.50 per GB for 1 TB vs. ∼\$0.15 per GB for 4 TB

SSD Architecture



Practical Issues

SSD performance has a tendency to degrade over time:

- Files are stored as a set of pages, typically 4 KiB in length.
- The pages are typically not located contiguously inside the device.
- Though, flash memory is accessed in blocks, typically 512 KiB in size.
- Consider a write to a page:
  1. The entire block where it is located must be read into a RAM buffer.
  2. The write is performed in the buffer.
  3. The entire block is erased in the flash memory (we can't do it in smaller chunks).
  4. The block from the buffer is written back into the flash memory.
- When the device gets full, files tend to be separated over more and more blocks.

Practical Issues

Counter measures:

- Over provisioning:
  - A substantial portion of the memory in the SSD is set aside as extra space for write operations.
  - Idle time is then used to defragment the stored data.
- TRIM command:
  - Allows an OS to inform the SSD which blocks are no longer considered in use, and can thus be wiped internally.

Flash memory becomes unusable over time after many writes:

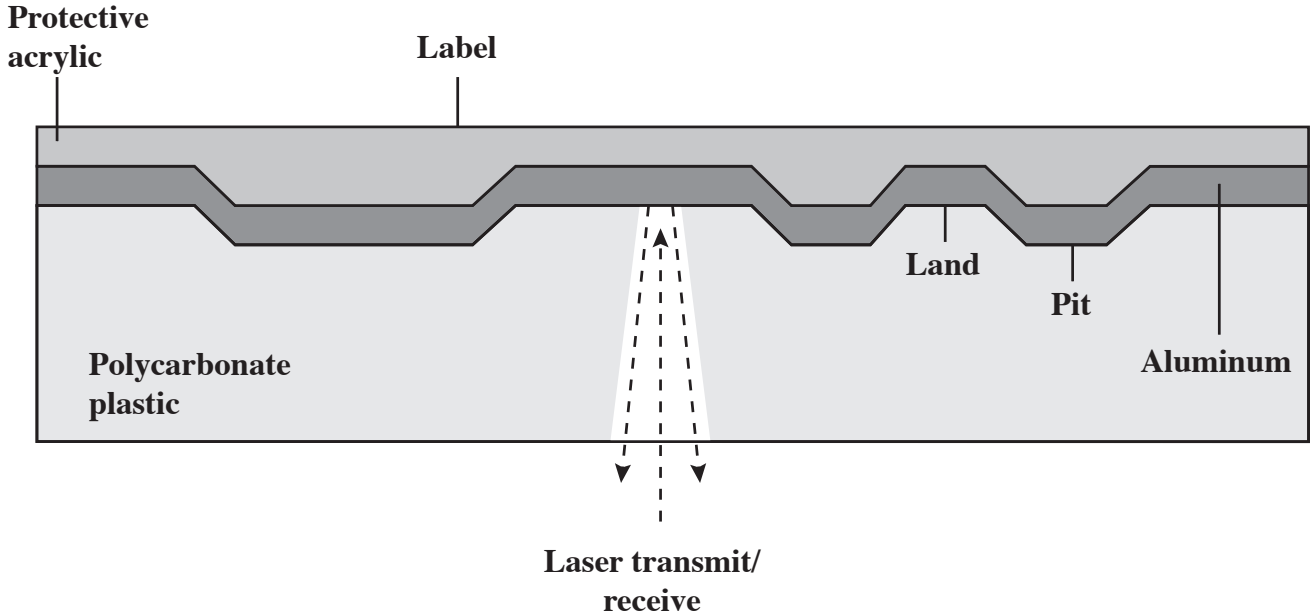
- Typical limit is 100 000 writes.
- Use a cache internally to delay and group write operations.
- Use wear-levelling algorithms to evenly distribute writes across blocks.
- Bad-block management techniques.
- Self estimation of own remaining lifetime to anticipate drive failure.

- Internal Memory
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  - Optical Discs
- Error Correction Codes

Overview

- CD-ROM / -R / -RW
  - Compact Disc Read-Only Memory / Recordable / Rewritable
  - Invented in 1983 by Philips and Sony.
  - Stores up to 800 MiB.
  - 12 cm diameter.
- DVD / -R / -RW
  - Digital Versatile Disk
  - Stores up to 17 GiB (double-sided DVDs).
  - 8 cm or 12 cm diameter.
- Blue-Ray Disc
  - Uses a 405 nm laser (blue-violet)
  - Stores up to 25 GiB in a single layer.

Principle of Optical Discs

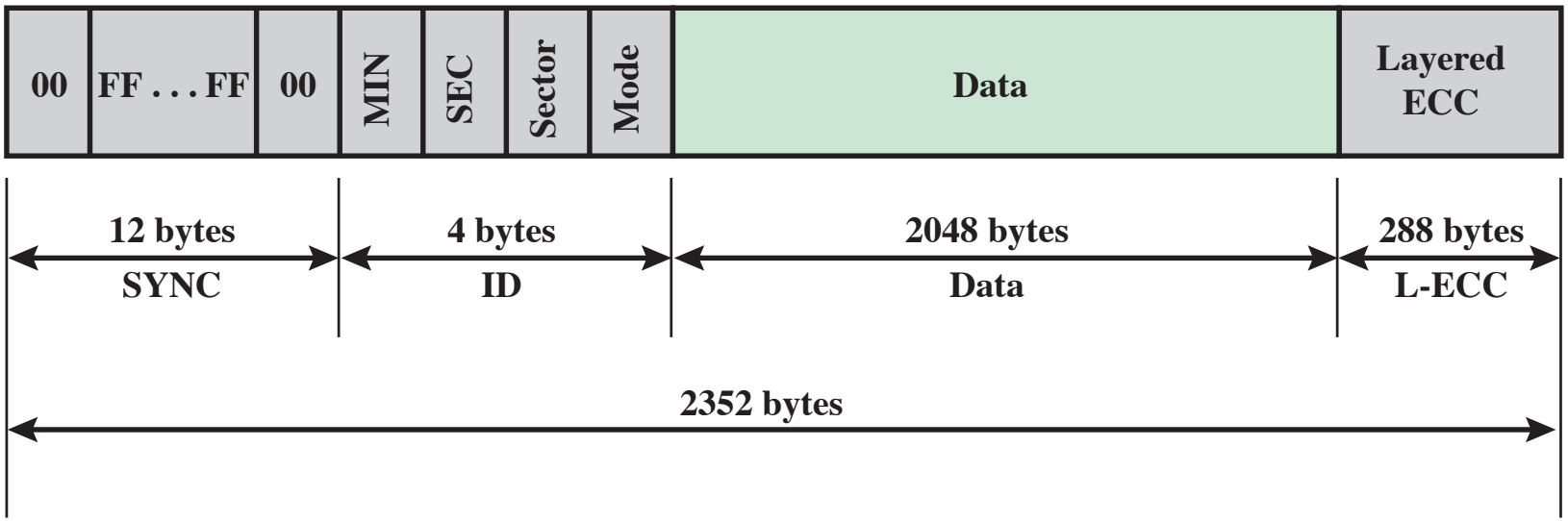


- The intensity of the reflected laser changes as it encounters a pit.
- A pit has a rough surface which scatters the light, i.e., a low intensity is reflected.
- A land has a smooth surface which reflects the light back with high intensity.
- A change in elevation means 1.
- No change in elevation means 0.

Principle of Optical Disks

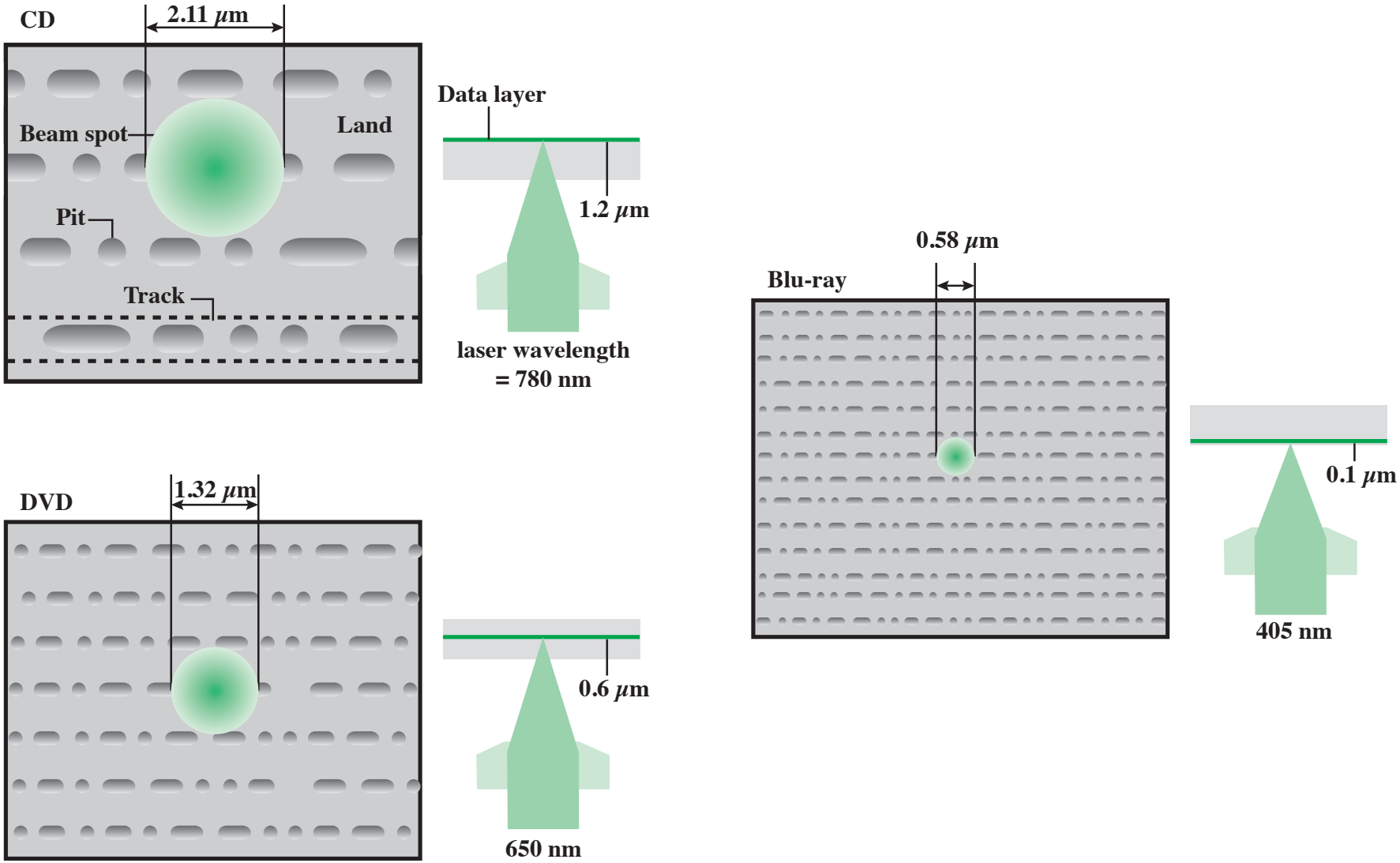
- Recall: on a magnetic disk, information is recorded in concentric tracks.
  - Either constant angular velocity (CAV), or
  - multiple zone recording.
- CDs use a single spiral track, beginning near the centre.
- All sectors have the same length.
- Information is packed evenly across the disk.
- Scanned at the same rate by rotating the disk at a variable speed.
- The pits are then read by the laser at a constant linear velocity (CLV).

A CD-ROM Sector



- Sync: identifies the beginning of a block.
- Header: contains the block address and the mode.
- Mode:
  - 0: a black data field.
  - 1: 2048 bytes of data + error-correcting code.
  - 2: 2336 byte of data without error-correcting code.

Optical Memory Characteristics



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  - Optical Discs
- Error Correction Codes

Memory Errors

- Hard failure
  - Permanent physical defect.
  - Memory cell(s) affected can not reliably store data.
  - They may be stuck at 0 or 1, or may switch erratically.
  - Can be caused by
    - Harsh environmental abuse
    - Manufacturing defects
    - Wear
- Soft error
  - Random, non-destructive event that alters the contents of one or more cells.
  - Can be caused by
    - Power supply problems
    - Alpha particles from radioactive decay

Most modern memory systems can have systems for detecting and correcting errors.

Error-Correcting Code Function

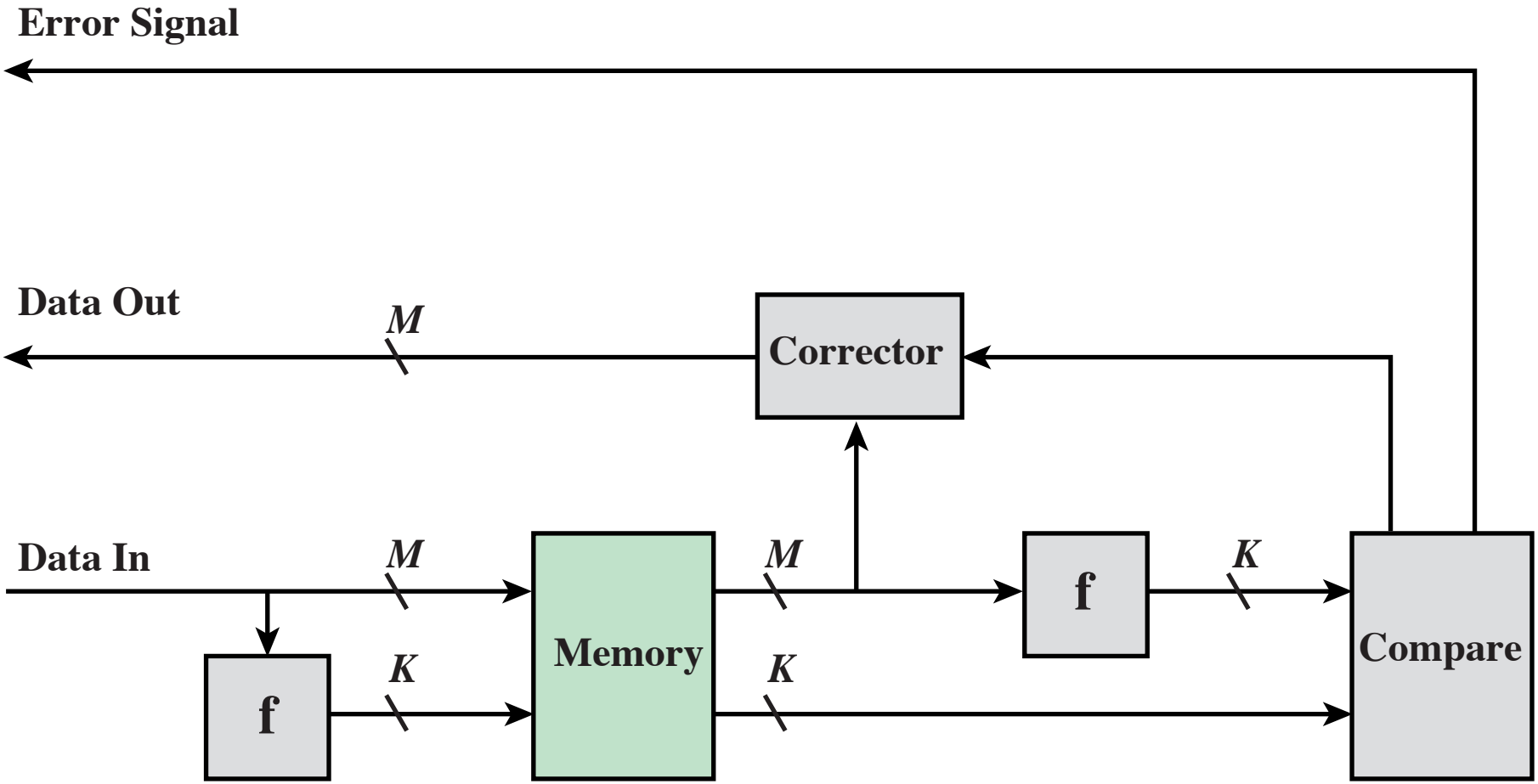


Figure 6.7 Error-Correcting Code Function

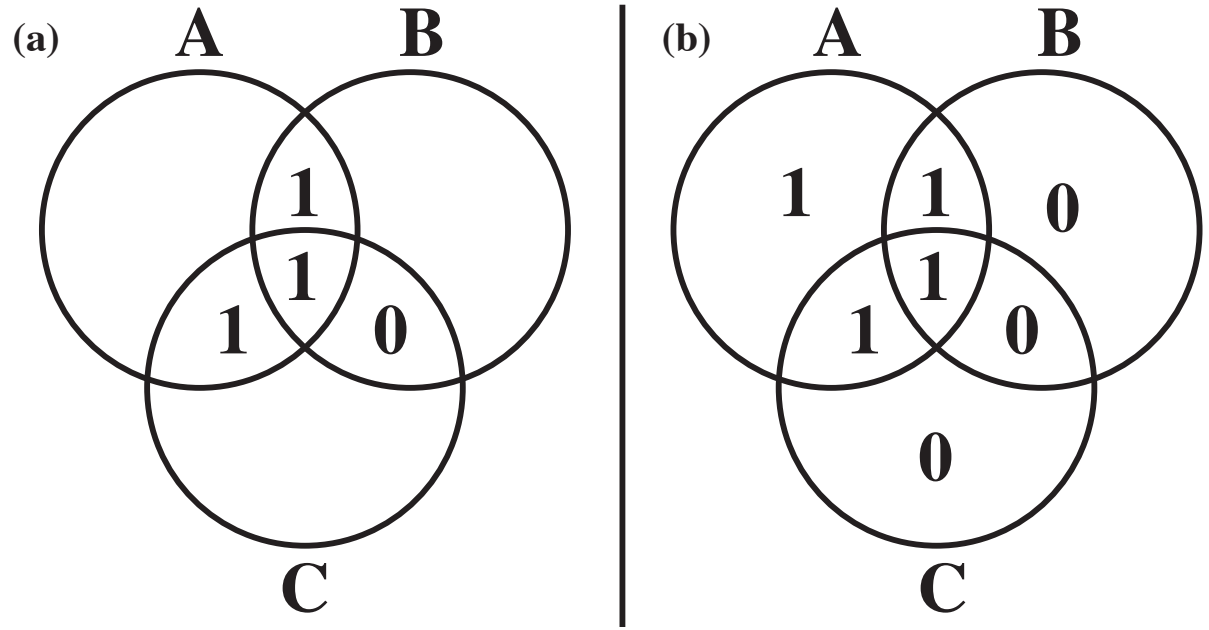
General Idea

- Given m data bits, assume for now that at most 1 bit is erroneously changed.
- To detect the change, we could add a single extra bit to make even parity.
- How about correcting the error? How many bits do we need?
- Add k bits and call them the check bits.
- In a sense we need enough to encode a position, including the added bits.
- And we also need "a pattern" to encode that there is no error.
- That is $m + k + 1 \leq 2^k$ must hold.
- E.g., for $m = 4$ we need $k \geq 3$.
- E.g., for $m = 8$ we need $k \geq 4$.
- E.g., for $m = 64$ we need $k \geq 7$.

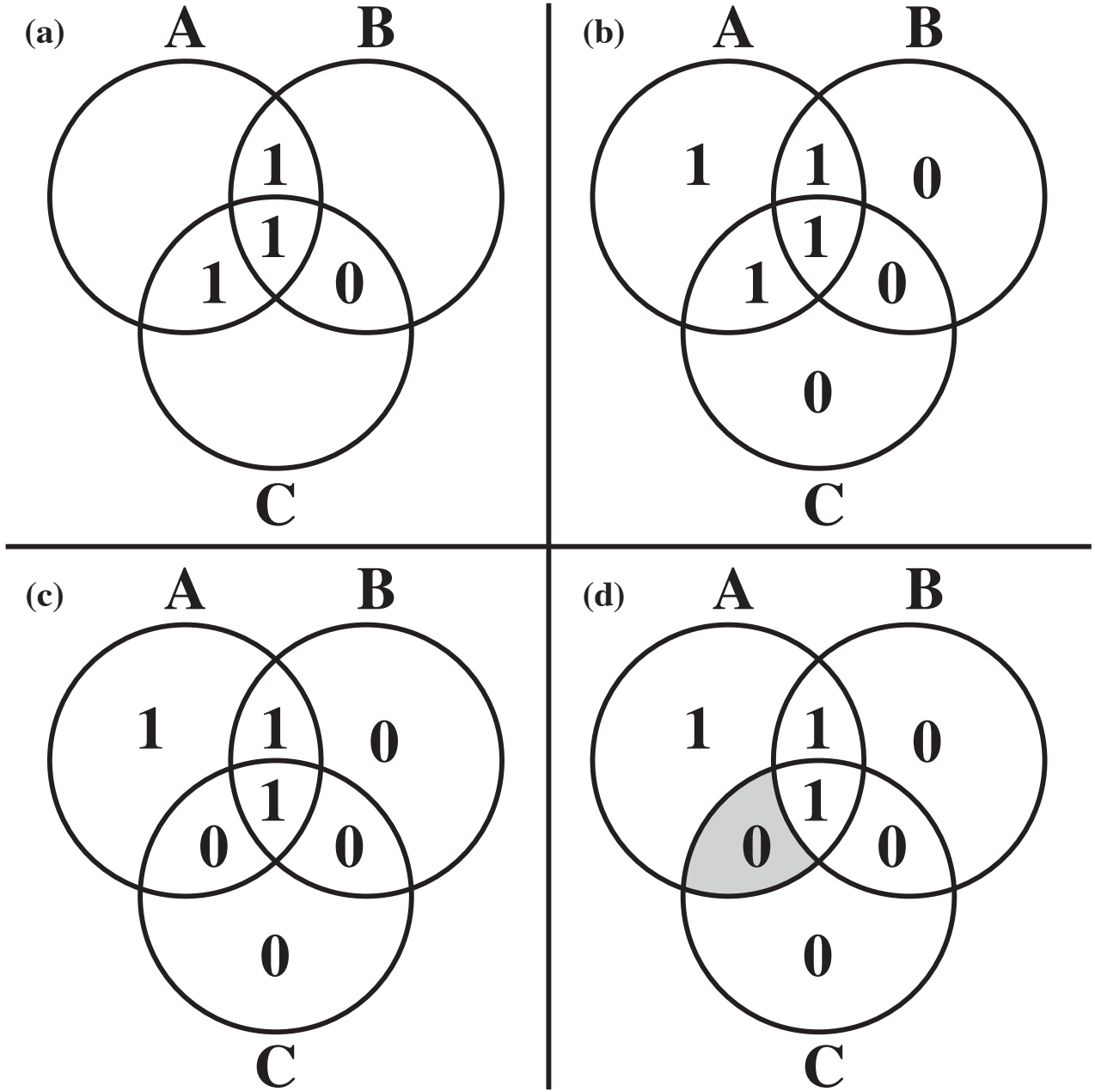
Hamming codes

- Devised by Richard Hamming at Bells Labs.
- Multiple check bits that have parity with different sets of bits.
- The check bits that are wrong encode the error position.

Abstract Error Correction with Hamming Codes



Abstract Error Correction with Hamming Codes



Hamming Code

Bit position	1	2	3	4	5	6	7
and in binary	001	010	011	100	101	110	111
Content							

- Designate each position as either data or check bit.
- Each check bit should check parity with half of the $n + k$ bits.

Hamming Code

Bit position	1	2	3	4	5	6	7
and in binary	001	010	011	100	101	110	111
Content	c_1						
c_1 checks	×		×		×		×

- Now, if c_1 is wrong, we know the error is in position 1, 3, 5, or 7.
- But an error in position 2, 4, or 6 goes unnoticed.

Hamming Code

Bit position	1	2	3	4	5	6	7
and in binary	001	010	011	100	101	110	111
Content	c_1	c_2					
c_1 checks	×		×		×		×
c_2 checks		×	×			×	×

- Now, if only c_1 is wrong, we know the error is in position 1 or 5.
- Each combination of right/wrong for c_1 and c_2 gives two possible positions (including "no error").

Hamming Code

Bit position	1	2	3	4	5	6	7
and in binary	001	010	011	100	101	110	111
Content	c_1	c_2	d_1	c_3	d_2	d_3	d_4
c_1 checks	×		×		×		×
c_2 checks		×	×			×	×
c_3 checks				×	×	×	×

- Each position is checked by a unique pattern of check bits.
- Given a code, calculate the syndrome (with $\oplus$ being XOR):

$$s_1 = c_1 \oplus d_1 \oplus d_2 \oplus d_4$$
$$s_2 = c_2 \oplus d_1 \oplus d_3 \oplus d_4$$
$$s_3 = c_3 \oplus d_2 \oplus d_3 \oplus d_4$$

- $s_3 s_2 s_1$ is then the position of the error, with 0 meaning "no error".
- The pattern can be extended similarly to larger (m, k).

Hamming Code

Bit position	1	2	3	4	5	6	7
and in binary	001	010	011	100	101	110	111
Content	c_1	c_2	d_1	c_3	d_2	d_3	d_4
c_1 checks	×		×		×		×
c_2 checks		×	×			×	×
c_3 checks				×	×	×	×

- Each position is checked by a unique pattern of check bits.
- Given a code, calculate the syndrome (with $\oplus$ being XOR):

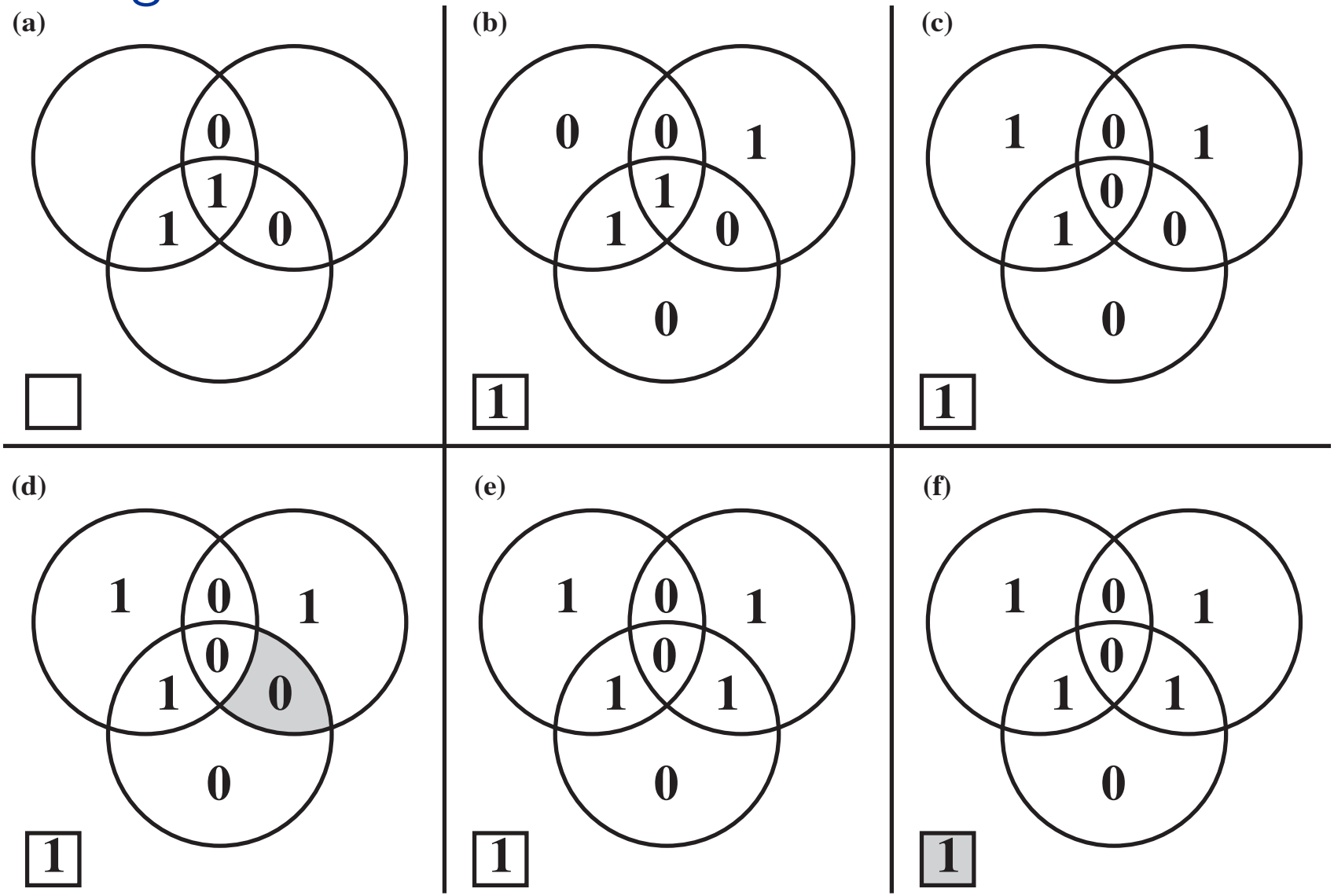
$$s_1 = c_1 \oplus d_1 \oplus d_2 \oplus d_4$$
$$s_2 = c_2 \oplus d_1 \oplus d_3 \oplus d_4$$
$$s_3 = c_3 \oplus d_2 \oplus d_3 \oplus d_4$$

- $s_3 s_2 s_1$ is then the position of the error, with 0 meaning "no error".
- The pattern can be extended similarly to larger (m, k).

Notes

- We can also use hypercubes to visualize what is going on (see supplemental material).
- Hamming codes are an simple example of a Single-Error-Correcting (SEC) code.
- Hamming codes are very easy to calculate.
- What if more than 1 error is introduced?
  - Either the errors are not detected, or
  - it is detected wrongly as a third error which is then "corrected".
- Single-Error-Correcting, Double-Error-Detecting (SEC-DED) code
  - Add a single extra bit to the Hamming code.
  - Generally considered a good trade-off between overhead and additional security.

Hamming SEC-DED Code



Space Overhead

	SEC		SEC-DED	
m	k	Increase (%)	k	Increase (%)
8	4	50.0	5	62.5
16	5	31.25	6	37.5
32	6	18.75	7	21.875
64	7	10.94	8	12.5
128	8	6.25	9	7.03
256	9	3.52	10	3.91

- DDR SDRAM can be bought with error correction (ECC)
  - Usually each rank then has 9 chips instead of 8.
  - That gives $m = 64$ and $k = 8$, enough for a Hamming SEC-DED code.